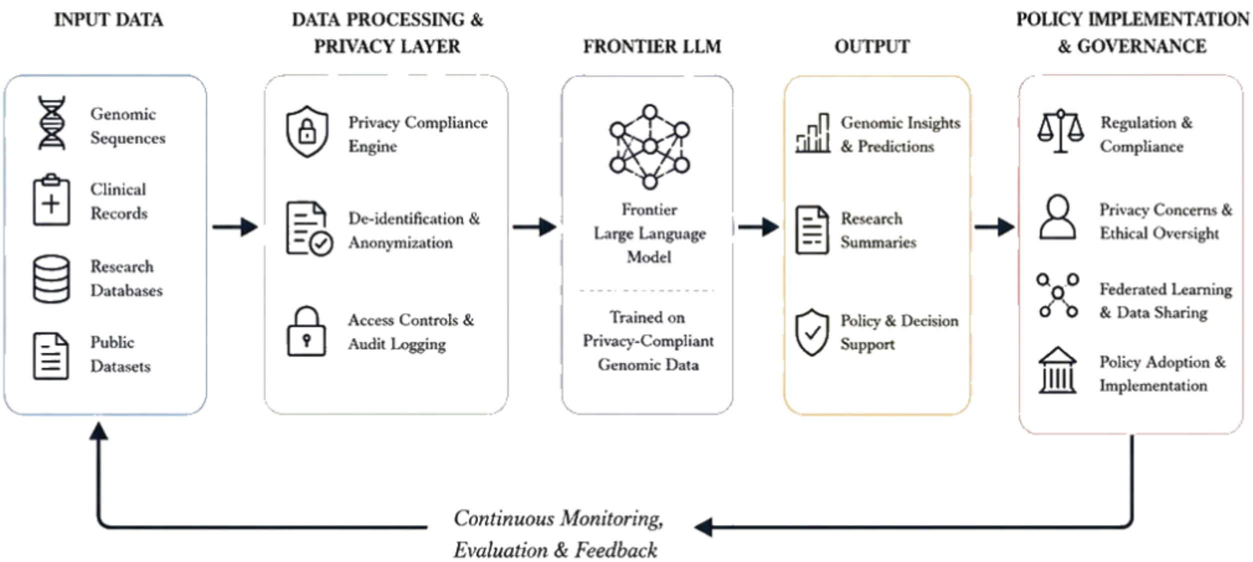

A Comprehensive Assessment of Genomic Data Privacy Regulations and Governance under Frontier LLM Use

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Executive Summary

This policy brief examines the developments and risks associated with the increasing frequency of utilization of artificial intelligence-powered architectures and methodologies within genomic analysis techniques. Our analysis culminates in a series of targeted policy recommendations for the Commonwealth of Massachusetts, particularly pertaining towards the manner in which Large Language Model (LLM) capability expansion may be coupled with an approach that prioritizes responsible data governance frameworks and architectures, as well as a greater degree of conversation between model deployment paradigms and hospitals themselves, in order to better protect patient and user privacy.

The field of genomic analysis via LLM approaches has progressed through several distinct phases, from early convolutional neural networks, which extracted localized sequence motifs with limited receptive fields, through hybrid architectures, such as Enformer, that were designed to model long-range genomic dependencies, to present-day large transformer foundation models including Nucleotide Transformer and DNABERT-2 that achieve genomic scale through self-supervised pretraining on large multi-species corpora. Each such phase has expanded the fundamental representational capacity that is available to investigators, doctors, and patients alike while simultaneously decreasing the degree of task-specific supervision required. Therefore, the interpretive bottleneck that historically constrained genomic analysis has been substantially, if not yet completely, mitigated; we project that this bottleneck will continue to shrink, at a differentially increasing rate, as model capabilities propagate and agentic systems absorb many such tasks.

This evolution of model capabilities has, in particular, led to the development of incredible analytical capacity, particularly within a capacity for identification of correlative and associative tendencies within long-range sequence data in broad subsets of genomes across organisms. Such capabilities have been demonstrated concretely by models in the EVE family, culminating in popEVE. The latter ranks mutations across the entire human proteome by synthesizing evolutionary sequence data with population-level resources including UK Biobank and gnomAD. In clinical validation against more than 31,000 families affected by serious developmental disorders, popEVE correctly identified the causal variant as the most damaging in the child's genome in 98% of cases where the causal mutation was already known, outperforming DeepMind's AlphaMissense and flagging only 11% of individuals as carrying severe variants at comparable thresholds, as opposed to AlphaMissense's 44%. Hence, it is clear that such models offer dramatic *potential* for significant upside impacts with respect to deployment in genome-wide association studies, particularly from the standpoint of ameliorating the impacts faced by patients with substantial disorders.

However, the rapid deployment of such capable systems introduces various privacy threats, data distribution concerns, and equity concerns that are in equal parts *substantial* as well as novel from a structural and implementation-based standpoint. The nature of genomic data itself is that (i) it offers an inherent identifiability, such that it may be distinguished in a categorical manner from other forms of information that may also be sensitive, (ii) it also may *link* individuals to those of their relatives, particularly given homology data and increased cosine similarity, and (iii) it encodes information about ancestry, disease predisposition, and physical characteristics that extends far beyond what any individual discloses intentionally. Traditional de-identification techniques are insufficient in this context, as sophisticated re-identification methods, such as data linkage across public databases, phenotype inference from sequence data, and pedigree reconstruction, may often compromise individuals and their non-consenting relatives even from datasets that have been ostensibly anonymized. That is, the differentiable *capabilities* of such models have outpaced, to orders of magnitude, the relevant privacy and safety considerations. Moreover, the deployment of genomic foundation models (FM's) in clinical settings introduces additional risks specific to the architecture of such models, including (i) the opacity of modern inference and pretraining pathways that make accountability processes exponentially more challenging (ii) the potential for training data memorization that could expose sensitive sequence information, particularly given reinforcement learning (RL) methodologies and (iii) extended context windows that characterize state-of-the-art models and substantially increase the probability of unintended data exposure.

Current federal and Massachusetts legal frameworks are inadequate to govern these risks. HIPAA's de-identification standards were designed for a pre-genomic regulatory environment and do not account for the re-identification capabilities of modern computational methods. The Genetic Information Nondiscrimination Act, moreover, provides protections against discrimination in health insurance and employment but leaves life, disability, and long-term care insurance entirely unregulated, which effectively creates gaps through which genomic information can be used adversely against individuals. Massachusetts Bill S.2619, which represents the most proximate legislative opportunity for state-level intervention, requires amendment to address these gaps comprehensively; moreover, *other* statewide bills address portions of the issues, but, as we contend, fail to address the aforementioned swath of associated risks regarding the deployment of FM's and LLM's within genomic analysis.

To govern such a rapidly transitory domain, we propose the following policy framework that includes, in broad terms, the following five recommendations:

(1) **First**, Massachusetts ought to amend Bill S.2619 to enact a complete prohibition on genetic discrimination across all insurance sectors, closing the gaps left by federal law.

(2) **Second**, the Commonwealth ought to establish technical standards for genomic AI systems, specifically defining biologically-grounded compute thresholds and extended-context window limits that trigger mandatory model registration and auditing.

(3) **Third**, consent frameworks for genomic research and clinical testing ought to be amended to mandate enhanced familial risk disclosure, ensuring that individuals understand the implications of data collection for biological relatives who are not party to the consent process.

(4) **Fourth**, Federated Learning ought to be adopted as the default computational standard for state-funded multi-institutional genomic research, enabling collaborative analysis without centralizing raw genomic data, complemented by standardized privacy budgets for Differential Privacy implementation across participating institutions. Our hope is that this paradigm is particularly influential for future model development considerations.

(5) **Fifth**, new regulations should govern the deployment of patient-facing AI systems in genomic medicine, imposing requirements for demonstrated clinical validity, robustness against confounding instrumental variables across diverse demographics and subsets of populations, as well as AI auditing of data transfer systems involved in genomic information.

Background – LLM Development and Utilization in Genomics

I. Framing Considerations.

The application of artificial intelligence (AI) methodologies to the tasks of genomic analysis, from sequence-wide studies to the resoluting phenomenological features, represents one of the most consequential interactions of computational and biological sciences within the past decade.[1] In particular, the genome itself comprises approximately three billion base pairs (hereafter abbreviated as *bp*), encoding the instructions for every protein, regulatory signal, and enzyme produced within the human body itself; the genome also accounts for a substantial fraction of the variation that distinguishes individuals in their susceptibility to disease itself, as well as their responses to treatments.[2,3] Even developmental trajectories may be affected given such information; therefore, the scale of this information has exceeded the capacity of traditional statistical methods to analyze in a comprehensive manner that simultaneously provides for high resolution and extract of endogeneity effects.[4] For instance, a genome-wide association study must, by specification, evaluate millions of variants while simultaneously controlling for various factors such as population structure and linkage disequilibrium.[5,6] This synthesis of high dimensionality and complex dependencies makes genomic analysis an unusually productive

[1] Eric S. Lander et al., "Initial Sequencing and Analysis of the Human Genome," *Nature* 409, no. 6822 (February 2001): 860–921.
[2] Ibid, p. 680–682.
[3] Teri A. Manolio et al., "Finding the Missing Heritability of Complex Diseases," *Nature* 461, no. 7265 (October 2009): 747–753.
[4] Peter Visscher et al., "10 Years of GWAS Discovery: Biology, Function, and Translation," *American Journal of Human Genetics* 101, no. 1 (July 2017): 5–22.
[5] David Heckerman, "A Bayesian Approach to Causal Discovery," in *Computation, Causation, and Discovery*, ed. Clark Glymour and Gregory F. Cooper (Cambridge, MA: MIT Press, 1999), 141–165. Cf. Alkes L. Price et al., "Principal Components Analysis Corrects for Stratification in Genome-Wide Association Studies," *Nature Genetics* 38, no. 8 (August 2006): 904–909.
[6] Peter M. Visscher et al., "Five Years of GWAS Discovery," *American Journal of Human Genetics* 90, no. 1 (January 2012): 7–24.

domain for the application of modern ML frameworks, particularly in the post-transformer series of models.[7] Such applications are particularly crucial for various *treatments* which are particularly common within Massachusetts itself, thereby emphasizing the necessity of models that are tailored to local needs.

It must be noted that the central challenge within seminal genomic analysis techniques is not the problem of *data scarcity* – indeed, following the conclusion of the Human Genome Project, most sequence-level data is available to base-pair accuracy – but rather the issue of data *complexity*. Indeed, an entire sequence that previously cost approximately 100 million dollars to attain in 2001 can now be produced for under one thousand dollars, such that the global volume of genomic data is estimated to double every seven months, improving the resulting time horizons.[8] Rather, the fundamental bottleneck for the investigative procedure of developing emergent results given sequence-level data (or even morphological or cosine-similarity data on proteins themselves) concerns the *interpretative* capabilities.[9] For one, raw sequence data must be aligned to a reference genome for systematic comparison ability; similarly, regulatory elements, such as enhancers, are incredibly difficult to identify without careful identifications and functional consequences described.[10,11] Each such step involves inferential problems of substantial difficulty, such that classical bioinformatic pipelines, which address these problems

[7] Alkes L. Price et al., "Principal Components Analysis Corrects for Stratification in Genome-Wide Association Studies," *Nature Genetics* 38, no. 8 (August 2006): 904–909.
[8] National Human Genome Research Institute, "DNA Sequencing Costs: Data," NHGRI Genome Sequencing Program, 2022, https://www.genome.gov/about-genomics/fact-sheets/Sequencing-Human-Genome-cost.
[9] Zachary D. Stephens et al., "Big Data: Astronomical or Genomical?," *PLOS Biology* 13, no. 7 (July 2015): e1002195.
[10] Heng Li and Richard Durbin, "Fast and Accurate Short Read Alignment with Burrows-Wheeler Aligner," *Bioinformatics* 25, no. 14 (July 2009): 1754–1760.
[11] Len A. Pennacchio et al., "Enhancers: Five Essential Questions," *Nature Reviews Genetics* 14, no. 4 (April 2013): 288–295.

through statistical models and empirical heuristics, are limited in their capacity to learn representations and generalize across heterogeneous contexts.[12,13]

Methodologies that leverage AI, and in particular deep learning as well as attention mechanisms, combat this limitation by effectively learning hierarchical representations of *sequence data* without requiring researchers to specify *in advance* which features are likely to be relevant, such that the dimensionality reduction problem becomes substantially simpler.[14] For instance, a convolutional model trained on genomic sequences does not require the investigator at hand to label the set of transcription factor binding motifs, particularly given transformer methods, or the interaction structure among distant genomic elements (or linked genes) to be explicitly modeled; rather, the model learns such representations from the fundamental set of training data, subject to the relevant inductive biases that are present within its architecture.[15] The capacity for such feature learning is especially valuable in genomics, particularly beyond traditional feature selection methods, such that the relevant biological signals may be isolated within an extremely large sample space that grows to exponential order in sequence length.[16]

The development of architectures tailored towards applications of developing emergent results based on genomic sequence characterizations has progressed through distinct phases,

[12] Babak Alipanahi et al., "Predicting the Sequence Specificities of DNA- and RNA-Binding Proteins by Deep Learning," *Nature Biotechnology* 33, no. 8 (August 2015): 831–838.
[13] Brendan J. Frey and Dustin Schuurmans, "Graphical Models for Machine Learning and Digital Communication," *IEEE Signal Processing Magazine* 15, no. 6 (November 1998): 37–50.
[14] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton, "Deep Learning," *Nature* 521, no. 7553 (May 2015): 436–444.
[15] Alipanahi et al., "Predicting the Sequence Specificities of DNA- and RNA-Binding Proteins by Deep Learning," *Nature Biotechnolog*y 33, no. 8 (2015): 831–838.
[16] David R. Kelley, Jasper Snoek, and John L. Rinn,"Basset: Learning the Regulatory Code of the Accessible Genome with Deep Convolutional Neural Networks," *Genome Research* 26, no. 7 (July 2016): 990–999.

matching (with some latency) the progression through classes of models within the broader ML community. Indeed, each such phase in the aforementioned progression has exhibited association with a class of models whose structural properties are well-suited to particular aspects of the genomic analysis problem, with decreasing degrees of (i) specificity required on part of the investigator with respect to inputs and outputs, and (ii) leverage on part of the investigator with respect to model architecture and training data considerations.[17]

Convolutional neural networks, which process sequences by applying learned filters across local windows of input while leveraging convolutions to extract features *without* requiring sequential recurrence relations, were among the first such architectures to demonstrate high performance on such prediction benchmarks.[18] In particular, the capacity among CNN's to detect local sequence motifs made them natural candidates for problems such as transcription factor binding prediction and chromatin accessibility modeling.[19] More recently, however, transformer-based models (originally developed for natural language processing and adapted to biological sequence analysis) have demonstrated the capacity to model dependencies and linkage relations, in an effective hierarchical manner, across the full length of a genomic sequence, demonstrating the ability to capture long-range regulatory interactions (and lack of such interactions thereof) that previous models could not represent in a compact manner.[20,21] Our analysis emphasizes that each such architectural development ought to be understood not as a

[17] Jesse Vig et al., "BERTology Meets Biology," *Proceedings of ICLR* (2021).
[18] Jian Zhou and Olga G. Troyanskaya, "Predicting Effects of Noncoding Variants with Deep Sequence-Based Neural Networks," *Nature Methods* 12, no. 10 (October 2015): 931–934
[19] Cf. Aliphani et al. (2015).
[20] Ji et al., "DNABERT: Pre-trained Bidirectional Encoder Representations from Transformers Model for DNA-Language in Genome," *Bioinformatics* 37, no. 15 (August 2021): 2112–2120.
[21] Žiga Avsec et al., "Effective Gene Expression Prediction from Sequence by Integrating Long-Range Interactions," *Nature Methods* 18, no. 10 (October 2021): 1196–120

replacement of its predecessors but as an extension of the representational capacity available to the relevant investigator, expanding the range of biological questions that can be addressed through computational means. The sections that follow describe each of these architectural families in turn, with attention to their structural properties and their representative application in genomics research.

II. Usage of Convolutional Neural Networks

Early deep learning applications in genomics relied on convolutional neural networks.[22] Models such as DeepBind, DeepSEA, and Basset applied one-dimensional convolutions across input DNA sequences.[23,24] These systems typically processed sequences ranging from several hundred to one thousand base pairs.[25] In a convolutional architecture, the model applies filters that slide across the sequence.[26] Each filter thereby learns the detection of a specific local pattern; in genomic contexts, such filters often *converge iteratively* towards representations of transcription factor binding motifs.[27] After convolution, models apply nonlinear activation functions and pooling layers; the pooling process reduces dimensionality in a manner that is functionally similar to Principal Component Analysis (PCA), yet far more scalable, and additionally aggregates local features into higher-level representations.[28] Fully connected layers

[22] LeCun, Bengio, and Hinton, "Deep Learning," *Nature* 521, no. 7553 (May 2015): 436–444.
[23] Alipanahi et al., "Predicting the Sequence Specificities of DNA- and RNA-Binding Proteins by Deep Learning," *Nature Biotechnology* 33, no. 8 (2015): 831–838
[24] Zhou and Troyanskaya, "Predicting Effects of Noncoding Variants with Deep Sequence-Based Neural Networks," *Nature Methods* 12, no. 10 (2015): 931–934
[25] Alipanahi et al. (2015); Zhou and Troyanskaya (2015)
[26] Kelley, Snoek, and Rinn, "Basset: Learning the Regulatory Code of the Accessible Genome with Deep Convolutional Neural Networks," *Genome Research* 26, no. 7 (2016): 990–999 (Basset).
[27] LeCun, Bengio, and Hinton, "Deep Learning," Nature 521, no. 7553 (May 2015): 436–444.
[28] Kelley, Snoek, and Rinn (2016).

then map these features to prediction outputs, such as binding probability or chromatin accessibility.[29,30] Such CNN-based models generally contained between one and ten million parameters; such models mandate the existence of supervised training, as well as training data derived from experiments such as ChIP-seq or DNase-seq.[31] Hence, the model receptive field remained limited by convolutional depth and filter width; as a result, such models captured local sequence motifs effectively but struggled to model interactions across tens of thousands of base pairs.[32]

An additional use of CNNs allows for better "image-based genomic data analysis, such as chromatin conformation capture (3C) data and microscopy images," and the use of Recurrent Neural Networks (RNN's) results in improvements in analysis of "sequential genomic data, such as DNA sequences and gene expression time-series data."[33,34] The performance of such algorithms effectively amounts to the process of poring through, and summarizing, the results and further extrapolative connections between (and from) thousands of pieces of scientific literature, electronic health records, and patient reports. From that information, the aforementioned natural language processing algorithms – in particular, CNN's and RNN's, together with more traditional methods, such as long short term memory models – can "automatically extract genetic variants, phenotypic data, and drug-gene interactions [which enables] efficient data integration, and knowledge discovery."[35] As a result of such applications

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[29] Alphanahi et al. (2015).
[30] Kelley, Snoek, and Rinn (2016).
[31] Zhou and Troyanskaya (2015); Kelley, Snoek, and Rinn (2016).
[32] Avsec et al., "Effective Gene Expression Prediction from Sequence by Integrating Long-Range Interactions," *Nature Methods* 18, no. 10 (2021): 1196–1203.
[33] Alfadhel, "Artificial Intelligence in Genomic Analysis."
[34] Jian Ma et al., "DeepC: Predicting 3D Genome Folding Using Megabase-Scale Transfer Learning," Nature Methods 17, no. 11 (2020): 1118–1124.
[35] Alfadhel, "Artificial Intelligence in Genomic Analysis."

and approaches of using AI, the main sub-fields where artificial intelligence are currently being applied are genomic data analysis, personalized medicine, diagnostics, and genome editing.[36,37] Within the former three subdisciplines, human expertise and errors can be accounted for by an automated artificial intelligence model, which allows for quicker and more accurate diagnoses for individual patients. With regards to genome editing, artificial intelligence is helping to optimize genome editing techniques such as CRISPR-Cas9 by predicting off-target effects and optimizing guide RNA design, which ensures safer and more efficient gene-editing outcomes to be applied to treatment of genetic disorders.[38] Unfortunately, some limiting factors to the ethics and access of artificial intelligence used in genomics are genomic data privacy and security of patients, informed consent, potential discrimination based on genetic information, and prohibitive costs of genetic treatments and personalized diagnoses.[39,40,41]

III. Hybrid Architectures and Long-Range Modeling

As genomic research increasingly emphasized regulatory interactions between distant elements, researchers developed architectures capable of modeling long-range dependencies. One prominent example is Enformer.[42] Enformer combines convolutional layers with transformer

[36] Chih-Hsuan Wei et al., "PubTator: A Web-based Text Mining Tool for Assisting Biocuration," *Nucleic Acids Research* 41, no. W1 (2013): W518–W522
[37] Bharath Ramsundar et al., *Deep Learning for the Life Sciences* (Sebastopol, CA: O'Reilly Media, 2019), chap. 4.
[38] Dara, Mahintaj, Mehdi Dianatpour, Negar Azarpira, and Nader Tanideh. “The Transformative Role of Artificial Intelligence in Genomics: Opportunities and Challenges.” Gene Reports 41 (December 2025): 102314. https://doi.org/10.1016/j.genrep.2025.102314.
[39] Alsaedi et al., “AI-Powered Precision Medicine.”
[40] Yann Joly et al., “Genomic Data Sharing and Privacy,” *Science* 378, no. 6619 (2022): 479–481
[41] Mark A. Rothstein, “Genetic Exceptionalism and Legislative Pragmatism,” *Hastings Center Report* 35, no. 4 (2005): 27–33
[42] Žiga Avsec et al., “Effective Gene Expression Prediction from Sequence by Integrating Long-Range Interactions,” Nature Methods 18, no. 10 (October 2021): 1196–203, https://doi.org/10.1038/s41592-021-01252-x.

attention layers. The model first applies convolutional blocks to extract local features from input DNA. It then feeds these features into transformer layers that implement self-attention. Self-attention computes pairwise relationships between positions in the sequence. Each position generates query, key, and value vectors. The model computes attention weights by comparing queries and keys. It then aggregates values according to these weights. This mechanism allows the network to integrate information from distant genomic regions.

Enformer processes input sequences of approximately 200,000 base pairs.[43] The model additionally predicts gene expression levels and other functional genomic signals.[44] The model contains on the order of hundreds of millions of parameters; its scale reflects both the depth of its convolutional backbone and the width and number of its attention layers.[45] By combining local feature extraction with global relational modeling, the architecture addresses limitations present in earlier CNN-only systems.[46]

The success of Enformer, in particular, within the process of integrating and long-range sequence information prompted subsequent research efforts to examine whether the *transformer* component uniquely (that operates directly on sequence representations without a convolutional preprocessing stage) could achieve comparable or superior performance to the collated model itself.[47] Such an investigation, and the resulting lack in the decline of performance, motivated the development of architectures that were purely attention-based, drawing in particular on advances in the natural language processing literature where transformer models operating on tokenized

[43] Avsec et al., "Effective Gene Expression Prediction from Sequence by Integrating Long-Range Interactions," *Nature Methods* 18, no. 10 (October 2021): 1196–1203. 196,608bp input length is specified in Methods.
[44] Avsec *et al.* (2021).
[45] Avsec *et al*. (2021).
[46] Avsec et al. (2021).
[47] Dalla-Torre et al., "The Nucleotide Transformer: Building and Evaluating Robust Foundation Models for Human Genomics," *Nature Methods* 21 (2024): 471–480.

input demonstrated strong generalization across broad categories of tasks that involved sequence-to-sequence comparative processing.[48]

Therefore, in large scale genomics settings, the analogous approach involves the process of tokenizing input DNA into subsequences of fixed-length, analogous to the *codon* phase processes within DNA translation itself.[49] Therefore, the embedding of each token into a continuous vector representation as well as the application of a multi-head self-attention mechanism across the sequence length produced models including Nucleotide Transformer and DNABERT-2.[50,51] Such models are thereby pretrained on large corpora of genomic sequences, functionally analogous to other state-of-the-art attention-based models, and thereby harnesses various masked language modeling objectives.[52] Such objectives allow the model itself to learn to predict masked tokens from their surrounding tokens; such a strategy during pretraining allows for the model to develop general-purpose representations of sequences that capture the properties of codon structure, cross-sequence interference patterns, and evolutionary conservation processes.[53]

Such a paradigm of generalization and iterative improvement within pretraining invokes a meaningful shift in the relationship between *model scale* and the fundamental requirements of

[48] Vaswani et al., “Attention Is All You Need,” *Advances in Neural Information Processing Systems* 30 (2017): 5998–6008
[49] Zhou et al., “DNABERT-2: Efficient Foundation Model and Benchmark for Multi-Species Genome,” *arXiv*:2306.15006 (2023)
[50] Dalla-Torre et al. (2024)
[51] Zhou et al. (2023)
[52] Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *Proceedings of NAACL* (2019): 4171–4186
[53] Elnaggar et al., “ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning,” *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, no. 10 (October 2022): 7112–7127

data.[54] As a re-emphasis, it is the *complexity* of such data that requires the considerations of model scale, whereas an increase in scale mandates the preprocessing of larger swaths of data, posing an iterative challenge. Consequently, whereas models based on CNN architectures, such as DeepSEA and Basset, required task-specific supervised training data derived from experiments such as ChIP-seq or DNase-seq, pretrained transformer models can be fine-tuned on comparatively small labeled datasets while retaining the representational richness acquired during pretraining.[55,56] The iterative preprocessing problem is therefore mitigated via such an architecture. Such an approach is therefore particularly valuable in genomics, where experimentally validated labels for specific regulatory phenomena remain expensive to generate and are often only available for specific cell types.[57] The ability to transfer representations learned from raw sequence data to prediction tasks therefore substantially expands the range of biological questions addressable through such ML methods, including promoter and enhancer identification, as well as the characterization of variants and substitutions in non-coding regions where annotation possibilities remain difficult.[58]

IV. Transformer Foundation Models for Genomics

More recent work applies pure transformer architectures to genomic sequence. DNABERT represents one of the earlier examples of this approach.[59] It adapts the BERT

[54] Kaplan et al., "Scaling Laws for Neural Language Models," *arXiv*:2001.08361 (2020).
[55] Zhou and Troyanskaya (2015); Kelley, Snoek, and Rinn (2016)
[56] Howard and Ruder, "Universal Language Model Fine-Tuning for Text Classification," *Proceedings of ACL* (2018): 328–339
[57] ENCODE Project Consortium, "An Integrated Encyclopedia of DNA Elements in the Human Genome," *Nature* 489, no. 7414
[58] Kircher et al., "A General Framework for Estimating the Relative Pathogenicity of Human Genetic Variants," *Nature Genetics* 46, no. 3 (March 2014): 310–315
[59] Yanrong Ji et al., "DNABERT: Pre-Trained Bidirectional Encoder Representations from Transformers Model for DNA-Language in Genome," Bioinformatics 37, no. 15 (August 2021): 2112–20, https://doi.org/10.1093/bioinformatics/btab083.

architecture from natural language processing. DNABERT tokenizes DNA into k-mers and trains using a masked language modeling objective.[60] During training, the model masks a subset of tokens and learns to predict them from the surrounding context. This objective encourages the network to learn contextual embeddings of genomic sequence. DNABERT models typically contain around 100 million parameters and train on the entire human reference genome.

Building on this paradigm, researchers have introduced larger genomic foundation models such as the Nucleotide Transformer. These systems scale parameter counts into the hundreds of millions or billions. Some variants contain up to several billion parameters. They train on thousands of genomes across species. Like large language models, they rely on stacked self-attention layers and feed-forward networks. Each transformer block includes multi-head attention mechanisms and position-wise fully connected layers. Residual connections and layer normalization stabilize training.

These foundation models treat genomic sequence as a large-scale unsupervised learning problem. After pretraining, researchers fine-tune them on downstream tasks such as promoter prediction, splice site identification, or variant effect estimation. Their scale enables them to learn generalized representations that transfer across tasks and organisms.

V. Comparative Trends in Architecture and Scale

Across the past decade, genomic AI has undergone three major methodological shifts. First, researchers deployed convolutional neural networks to detect local motifs within short sequence windows. These models contained millions of parameters and focused on supervised

[60] Ibid.

prediction tasks. Second, hybrid architectures incorporated attention mechanisms to capture long-range regulatory interactions. Parameter counts increased into the hundreds of millions, and input context expanded to hundreds of thousands of base pairs. Third, large transformer foundation models adopted self-supervised pretraining at genomic scale. These systems now contain hundreds of millions to billions of parameters and train on multi-species corpora.

In architectural terms, CNN-based models emphasize hierarchical feature extraction through localized filters. Transformer-based models emphasize pairwise relational modeling through attention matrices. Hybrid systems combine these approaches by using convolution to extract local structure before applying attention to integrate distant signals.In computational terms, model growth reflects increased data availability and hardware capability. Early genomic networks trained on experiment-specific datasets. Foundation models train on entire genomes and extensive public repositories. As parameter counts expand, models require distributed training across multiple accelerators. Taken together, state-of-the-art genomic AI methods rely on deep neural architectures that scale in both depth and width. They encode DNA either as one-hot vectors or as token embeddings. They process sequence through convolutional filters, self-attention layers, or combinations of both. Their parameter counts now rival those of mid-scale language models. Understanding these systems requires attention to their structure, their training objectives, and the magnitude of their computational footprint.

VI. State-of-the-art genomics models within Massachusetts: popEve and PDgrapher

Within Massachusetts, there are several notable research laboratories, especially those linked to prestigious schools such as Harvard or the Massachusetts Institute of Technology. The presence of these schools is significant when observing the progression of artificial intelligence use in genomics from a Massachusetts perspective because while the implications of the results from research performed at these schools is intended to be national, the preliminary application is usually done through large hospitals in Massachusetts. An example of how AI research in genomics is directly applied to Massachusetts hospitals is popEve. Developed by Harvard Medical School researchers, popEve is a new AI model that can accurately diagnose a patient's likelihood to have a genetic disease based on variations in their genome. Importantly for Massachusetts, while the AI learning model is available for global use, the team of researchers that created popEve are specifically "collaborating with organizations including the Children's Rare Disease Collaborative at Boston Children's Hospital." Since Massachusetts is at the forefront of innovation in AI use within genomics, it is also at the forefront of application of AI for treatment of patients, meaning that citizens of Massachusetts will be among the first to experience the benefits of AI in expediting and expanding the accessibility of previously complex methods of genomic analysis and personalized treatment. To demonstrate that the popEve is not an isolated case, there is another AI model developed by Harvard Medical School researchers known as PDGrapher that is able to identify the main factors causing diseases within cells and create possible therapies that reverse the deterioration of those cells. The implications of PDGrapher is a revolutionization in drug discovery, where new treatments can be discovered for previously incurable diseases like Alzheimer's or Parkinson's. Currently, that team of Harvard researchers is "collaborating with colleagues at the Center for XDP at Massachusetts General Hospital," again showcasing how the next step for research performed at prestigious schools like Harvard is to apply their findings to hospitals within the Massachusetts area.

These leading institutions perpetually remain at the forefront of advancing genomic technology, such as popEVE, a model developed in the Harvard Medical School's Blavatnik Institute by Professor Debora Marks et. al.[61] The popEVE model itself produces a score for each variant in a patient's genome, indicating the variant's likelihood of causing disease and thereafter places these variants on a continuous spectrum. Additionally, such a real-valued score function is included as a *superposition* of the various factors that are incorporated within the likelihood (or the log-likelihood thereof), with the parameters for such a superposition being optimized after pretraining and prior to the inference step.[62] The most notable aspect of popEVE is that it can predict whether variants are benign or pathogenic, and which variants are more likely to lead to death in childhood versus adulthood, also identifying more than 100 previously unknown alterations that were responsible for undiagnosed rare genetic disease.[63] In Massachusetts as opposed to other states within the U.S., there is an advantage of sitting in a clinical network, constantly being validated and piloted in Massachusetts hospitals – particularly given the *collaborative* network of such hospitals in the manner of research dissemination.[64] More importantly, popEVE stands out from other tools because it solves this "proteome-wide calibration" problem that perturbed other tools.[65] For instance, a variant that would look severe in one protein couldn't be accurately presented alongside a variant in another.[66] It is a problem doctors need to know which mutation in a patient's genome is the most damaging; this allows for *selective treatments* based on the mutations that are most likely to cause damaging diseases,

[61] Frazer et al., "Disease Variant Prediction with Deep Generative Models of Evolutionary Data," *Nature* 599, no. 7883 (November 2021): 91–95
[62] Frazer et al (2021).
[63] Spence et al., "Calibrated Variant Pathogenicity Prediction Across the Human Proteome" *Nature Genetics* (2024)
[64] Spence et al. (2024)
[65] Spence et al. (2024)
[66] Bycroft et al., "The UK Biobank Resource with Deep Phenotyping and Genomic Data," *Nature* 562, no. 7726

rather than approaches that may select for rather impotent mutations that result from a lack of reasonable filtration of high-signal results within large sample sizes (of single-nucleotide polymorphisms, for instance). Based on the latest model in the EVE family, popEVE thereby to tackle this problem through the means of combining evolutionary data with UK Biobank and gnomAD data to produce the first model that ranks mutations across the entire human proteome; such an approach may also be classified as an elementary federated learning strategy, which we discuss further in the recommendation section below.[67]

Down a few streets, the Broad and Dana-Farber joined forces to produce EpiBERT, an AI model which was inspired by BERT, a deep learning model designed to understand and generate human-like language that was trained on data gathered from a myriad of human cell types to predict which genes are expressed in any type of human cell.[68] This was done in tandem with the implementation of PUPs, a method for prediction of unseen protein's subcellular location, which combines a protein language model with image in-patient model to utilize protein sequences and and cellular images to assess and notify changes in localization due to mutations in protein sequences, providing insights into potential mechanisms of disease.[69]

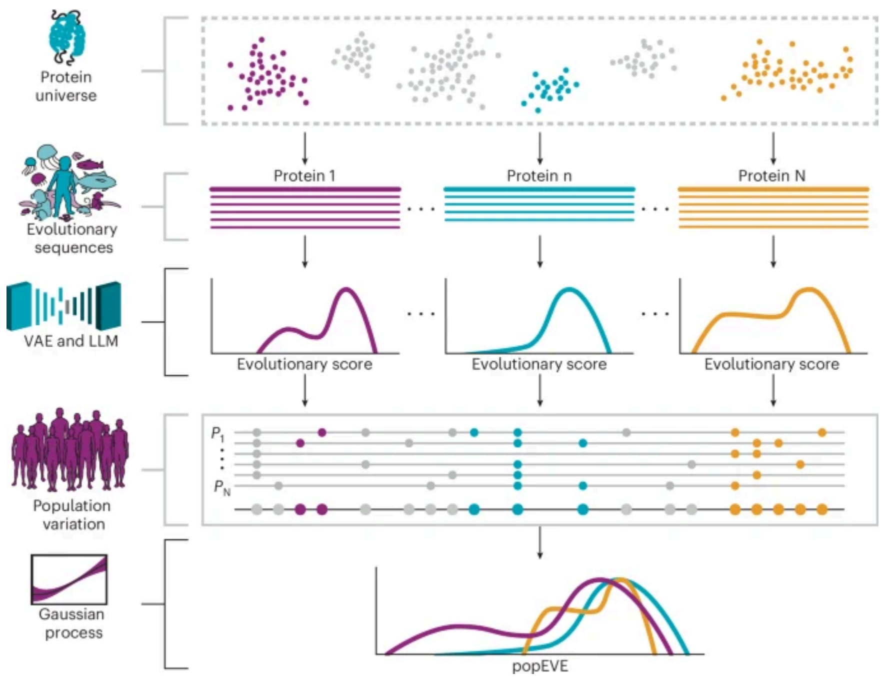
Regarding diagnostic accuracy, in order to validate popEVE, an analysis went underway for more than 31,000 families with children affected by serious developmental disorders.[70] For 98% of cases where a casual mutation was already identified, popEVE correctly ranked that

[67] Cf. Karczewski et al., "The Mutational Constraint Spectrum Quantified from Variation in 141,456 Humans," *Nature* 581, no. 7809 (May 2020): 434–443.
[68] Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *Proceedings of NAACL* (2019): 4171–4186
[69] Elnaggar et al., "ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, no. 10 (2022): 7112–7127
[70] Deciphering Developmental Disorders Study, "Large-Scale Discovery of Novel Genetic Causes of Developmental Disorders," *Nature* 519, no. 7542 (March 2015): 223–228.

variant as the most damaging in the child's genome.[71] Surprisingly, popEVE outperformed state-of-the-art competitors including DeepMind's AlphaMissense, which failed to reach the benchmark accuracy rate. AlphaMissense also provides significantly effective variant predictions, but popEVE takes this further to reduce false positive predictions within the general population (for instance, it flags only 11% of individuals as carrying severe variants at comparable thresholds as opposed to AlphaMissense's 44%).

Additionally, the popEVE framework is also predominantly accurate in assessing mutations in non-European populations.[72] popEVE recalls severe genetic disorder cases without overpredicting pathogenicity in the general population.This links to the safety feature that most prominently distinguishes popEVE, particularly its treatment of ancestral bias.[73] Considering the sector as a whole, that of broader genomics AI, most models are trained overwhelmingly on data from individuals of European ancestry.

[71] Spence *et. al*. (2024).
[72] Spence *et. al*. (2024).
[73] Spence *et al. (*2024).



***Above*. PopEve performance validation, as conducted by Spence et al. (2024).**

An analysis of the first decade of polygenic scoring studies found that 67% included purely European ancestry participants and another 19% only East Asian ancestry participants.[74] That left just 3.8% of studies pertaining to African, Hispanic, or Indigenous peoples, and because of this bias, other models are more likely to flag variants common in non-European populations as pathogenic simply because those variants are unfamiliar to the model.[75] With fewer false positives, popEVE exhibits significantly greater accuracy in avoiding bias in genetic variants incorrectly being classified as pathogenic. However, such models are still undergoing clinical validation processes, such that researchers themselves assert that more testing is needed before deployment. Beyond the necessity for accuracy within testing, various implications of *legal use* must be considered given the possibility for the exhibition of such inductive biases given

[74] Martin et al., "Clinical Use of Current Polygenic Risk Scores May Exacerbate Health Disparities," *Nature Genetics* 51, no. 4
[75] Spence *et. al.* (2024).

tendencies within model training data, particularly given emergent behaviors in genome-wide association studies that may exhibit strongly distinct implications for distinct demographics.

VII. Miscellaneous Model Development Considerations

Although Harvard Medical School seems to be the most common collaborator with Massachusetts hospitals regarding their research into AI use in genomics, there does not appear to be any reasons why other schools such as MIT would not or could not do the same. MIT researchers have already developed an AI learning model that can “optimize the genetic sequences of proteins…making production more efficient” and another AI model called ChromoGen that decreases the time necessary to decode the structural composition of a genome from days to minutes. Though there does not appear to be any attempts to expand the application of these innovations beyond sharing the models with the world, it seems likely based on the actions of other influential Massachusetts universities like Harvard that MIT would partner with Massachusetts hospitals if they were to apply their models to actual patient care.

Beyond innovations from universities, there has also been notable research done by local Massachusetts hospitals. Researchers at Mass General Brigham have expanded on the frontier of both radiomics and genomics through developing a machine learning algorithm that “quantifies brain age and shows potential for predicting outcomes after stroke,” as well as a predictive model that assists researchers with “understanding the genomic status of glioblastoma tumors” to reliably perform molecular diagnosis of the tumors. Although their research is still at its beginnings, the researchers clearly express their desire to use their findings to help patients with brain tumors, demonstrating how AI use in genomics is already directly improving the lives of patients within Massachusetts. Something to consider, however, is that a significant portion of the research done at universities and large hospitals is concentrated in urban areas, meaning that the benefits of AI use in genomics may not be broadly applicable to rural residents of Massachusetts as well.

A potential application of such models, with significantly greater positive application possibilities than the downstream effects discussed in the latter portion of our analysis, involves the probability that several genetic disorders may be mitigated (or ameliorated) by such acceleratory techniques. Genetic disorders, broadly defined, are conditions caused by a mutation that affects one's genes, or otherwise has some change to their genetic material. Common examples include fragile X syndrome,[76] down syndrome,[77] and Klinefelter syndrome,[78] among many others. Genomic testing can be used to detect these diseases. However, genomic data can be used in various other ways to improve patient outcomes. For instance, Mass General offers genomic services spanning oncology, cardiovascular health, the endocrine system, gastroenterology, ophthalmology, and more. The field of genetics is rapidly expanding. The discovery of the double-helix structure of DNA was not discovered until the 1950s,[79] and the complete human genome was not sequenced until 2022.[80]

[76] Cleveland Clinic. "What Is Fragile X Syndrome (FXS)?" Accessed February 22, 2026. https://my.clevelandclinic.org/health/diseases/5476-fragile-x-syndrome.
[77] Cleveland Clinic. "Down Syndrome: Symptoms & Causes." Accessed February 22, 2026. https://my.clevelandclinic.org/health/diseases/17818-down-syndrome.
[78] Cleveland Clinic. "Klinefelter Syndrome." Accessed February 22, 2026. https://my.clevelandclinic.org/health/diseases/21116-klinefelter-syndrome.
[79]Francis Crick - Profiles in Science. "The Discovery of the Double Helix, 1951-1953." March 12, 2019. https://profiles.nlm.nih.gov/spotlight/sc/feature/doublehelix.
[80] Nurk, Sergey, Sergey Koren, Arang Rhie, Mikko Rautiainen, Andrey V. Bzikadze, Alla Mikheenko, Mitchell R. Vollger, et al. "The Complete Sequence of a Human Genome." Science 376, no. 6588 (April 2022): 44–53. https://doi.org/10.1126/science.abj6987.

Recent reviews showcase how AI models enhance genome-wide association studies (GWAS) through the process of automating analysis of a myriad of single-nucleotide polymorphism (SNP) datasets and using approaches like EigenGWAS and DeepWAS to uncover variants associated with complex disease; this thereby improves individual risk prediction and also helps support prevention strategies.[81] In clinical practice, AI-driven genomics helps physicians and other hospitalists in disease screening, early diagnostics, and treatment selection through the means of decoding genetic information and simultaneously integrating multi-omics (which comprises genomic, transcriptomic, and epigenetic data) with electronic health record data. Together, this multi-modal approach enables more personalized care plans. Ultimately, these tools are used all the way from rare disease diagnostics to pharmacogenomics, where AI can help interpret variants of unknown significance and also optimize drug choice based on patient genotype.

One domain in which these capabilities converge with particular clinical urgency is in prenatal and neonatal settings, where the early identification of genetic disorders can meaningfully alter both the trajectory of disease management and the scope of available interventions.[82] Whole-genome sequencing of newborns (a practice that is now being piloted at scale within Massachusetts specifically, through, for instance, the BabySeq project at Brigham and Women's Hospital) generates a volume of variant data that exceeds the interpretive capacity of clinical geneticists working without computational assistance.[83] Such interpretive capacities

[81] Dara, Mahintaj, et al. “The Transformative Role of Artificial Intelligence in Genomics: Opportunities and Challenges.” Gene Reports, vol. 41, Dec. 2025, p. 102314, https://doi.org/10.1016/j.genrep.2025.102314.
[82] Ceyhan-Birsoy et al., "Interpretation of Genomic Sequencing Results in Healthy and Ill Newborns: Results from the BabySeq Project," *American Journal of Human Genetics* 104, no. 1 (January 2019): 76–93.
[83] Ceyhan-Birsoy et al. (2019).

satisfy the aforementioned *feasibility*, as well as explainability, constraints that medical professionals typically seek for the relevant sources of input upon the decision-making process.[84,85]

Therefore, AI-driven variant prioritization tools, including those built on the EVE and popEVE frameworks, can reduce the interpretive burden by ranking variants according to predicted pathogenicity before a clinician reviews the case, thereby accelerating the diagnostic pathway for infants presenting with symptoms of unknown etiology.[86] The clinical stakes of this acceleration are substantial: for conditions such as spinal muscular atrophy, phenylketonuria, and certain inborn errors of metabolism, the window during which intervention is most effective is measured in weeks rather than months, such that a delay in diagnosis of even modest duration can produce irreversible developmental consequences.[87] Beyond the neonatal setting, AI models trained on multi-generational pedigree data are increasingly being deployed to estimate carrier status and recurrence risk for prospective parents, integrating population-level allele frequency data from resources such as gnomAD with family-specific variant information to produce individualized risk estimates that are substantially more precise than those derivable from population statistics alone.[88] The convergence of these capabilities – in particular, including prenatal risk stratification, neonatal variant prioritization, and carrier screening – indicates that the transformative near-term applications of AI in genomic medicine may lie not necessarily within the treatment of established disease but, rather, its prevention and earliest detection, a

[84] Spence et al. (2024)
[85] Rentzsch et al., "CADD: Predicting the Deleteriousness of Variants Throughout the Human Genome," *Nucleic Acids Research* 47 (January 2019): 886–894.
[86] Mercuri et al., "Spinal Muscular Atrophy," *Nature Reviews Disease Primers* 8, no. 1 (February 2022): 52
[87] Blau et al., "Phenylketonuria," *Lancet* 381, no. 9874 (April 2013): 1417–1427
[88] Nykamp et al., "Sherloc: A Comprehensive Refinement of the ACMG-AMP Variant Classification Criteria," *Genetics in Medicine* 19, no. 10 (October 2017): 1105–1117

reorientation of clinical genomics that carries significant implications for how healthcare systems allocate resources and structure genetic counseling services.[89]

[89] Khoury et al., "Transforming Public Health Genomics: Defining and Measuring Success," *Genetics in Medicine* 19, no. 12 (December 2017): 1263–1265.

Assessment of Threat Imminence

Unique Threats Posed to Privacy by LLM-fueled Genomic Analysis Techniques

Methods including data linkage, DNA phenotyping, pedigree analysis, genotype imputation, and phenotype inference pose potential risks by re-identifying de-identified genomic records by linking DNA data to outside datasets, inferring missing traits or relatives, or predicting visible or health-related characteristics that narrow a person's identity.[90] [91] Deep learning models can leak private information through gradient extraction during training or through attacks on trained models, and even protein expression levels from proteomics data could potentially identify individuals. Pedigree analysis attacks don't just compromise individual patients but reveal sensitive family health information, impacting relatives who never consented to data sharing, and DNA phenotyping enables the prediction of facial features, hair color, and skin color, making patients physically identifiable from their genetic data. [92] An example of this is the 2013 Gymrek et al. study, which showed that supposedly anonymous genomes could be re-identified by inferring surnames from Y-chromosome markers and combining them with basic metadata like age and state. [93]

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[90] Annan, et. al,. *Genomic Privacy and Secuirty in the Era of AI,* 3-10

[91] Zhou, Juexiao, Chao Huang, and Xin Gao. 2024. "Patient Privacy in AI-Driven Omics Methods." *Trends in Genetics* 40 (5): 383–86. https://doi.org/10.1016/j.tig.2024.03.004.

[92] Zhou, Chao, Xin, *Patient Privacy, 1-3*

[93] Gymrek, Melissa, Amy L. McGuire, David Golan, Eran Halperin, and Yaniv Erlich. 2013. "Identifying Personal Genomes by Surname Inference." Science 339 (6117): 321–24. https://doi.org/10.1126/science.1229566.

Once shared and trained for AI development, genetic information could resurface in harmful ways: patients asking to be anonymous may be reconnected to real identities, or a third party malicious actor could gain unauthorized access to sensitive information. Currently, external checks on evolving AI remain weak, highlighting the need for smarter state policies guarding genetic data, implementing internal checks in the algorithms, clearly defining legal responsibility when errors occur, and preventing biases in training sets to integrate into AI systems.

Transparency Framing: Fundamental Incompatibility

Although many, both in the medical field and outside, have recognized the benefits that AI brings to the field, there are equally as many hesitations surrounding its growth. A Per Research Center survey showed that 60% of their participants noted they would be "uncomfortable" if decisions by their healthcare provider were made using AI tools.[94] While multiple federal regulations have been put in place to govern AI usage, such as the Executive Order 14110, the U.S. Food and Drug Administration, the Centers for Medicare & Medicaid Services, and others, there are unresolved difficulties with AI tools, especially in healthcare.[95] One major problem is the "black box" issue, where AI can only be viewed in terms of what is input and output without clear knowledge of its algorithm.[96] This creates problems with full transparency in healthcare, which can make it difficult to determine who is at fault.

[94] Nora Wells, *Artificial Intelligence (AI) in Health Care*, CRS Report R48319 (Washington, DC: Congressional Research Service, December 30, 2024), accessed February 23, 2026, https://www.congress.gov/crs-product/R48319

[95] April J. Anderson, Paulette C. Morgan, Amanda K. Sarata, and Nora Wells, *Artificial Intelligence (AI) in Health Care*, CRS Report R48319 (Washington, DC: Congressional Research Service, December 30, 2024), accessed February 23, 2026, https://www.congress.gov/crs-product/R48319.

[96] Hanhui Xu and Kyle Michael James Shuttleworth, "Medical Artificial Intelligence and the Black Box Problem: A View Based on the Ethical Principle of 'Do No Harm'," Intelligent Medicine 4, no. 1 (2024): 52–57, https://doi.org/10.1016/j.imed.2023.08.001

Necessity of Reliability Estimation Techniques

To comparatively examine the reliability of AI genomic models developed in Massachusetts, there must first be a national average of the reliability of these models. A common issue found in machine learning algorithms with regards to identification of risk factors from genomic data is that these models can mistakenly link several genetic variations to an individual's risk of developing a genetic disease, even though several of those variables may be unrelated. The complication here is likely due to the complex interactions between genetic factors, making it difficult to pinpoint the exact variation that causes the disease. But while this issue appears to be a standard issue for most AI models, specific models developed in Massachusetts like popEve "did not show ancestry bias by performing worse in people from underrepresented genetic backgrounds and did not overpredict the prevalence of pathogenic variants." By maintaining the same level of accuracy in identification even when faced with underrepresented genetic backgrounds, popEve demonstrated that it did not mistakenly link unrelated genetic variations to diseases since it did not overpredict the prevalence of the diseases from its identification of risk factors, implying a high level of reliability and accuracy. It is more difficult to examine the comparative safety of AI models developed in Massachusetts compared to the nation because researchers do not usually disclose how vulnerable their models are to misuse of patients' genomic data, so comparing these specific models is relatively unfeasible at this current moment. Even without specific comparisons, however, there are new methods that improve on protection of patient privacy.

Necessity of Privacy Maintenance coupled with Model Development

As genomic analysis using AI develops, companies and research organizations face vulnerability to cyber threats that can steal and misuse patients' genomic information. When the genomic data is stored through third-party cloud services, it becomes especially susceptible to unauthorized access if that platform's cybersecurity is compromised. A method to increase protection of individual privacy is differential privacy which inserts a level of randomness into the data to ensure that an individual's specific data is difficult to trace among all of the data.[97] There are issues of efficiency regarding utilization of individual data for personalized diagnoses, however, it can be adjusted using other methods such as cryptographic models, which gather information without having to decrypt encrypted data, to increase data utility while still increasing privacy protection overall. Another method to decrease risks of data breaches is federated learning, which trains the AI models using data from multiple decentralized sources before combining the models to create a global model.[98]

By decentralizing the data, cyber attackers are limited in the amount of sensitive data they can access while the AI model is still trained to handle a variety of genomic analysis tasks. But despite an increase in direct privacy protection, federated learning has a flaw due to its vulnerability to membership inference and model inversion attacks, which analyze the behavior of the model to determine whether a specific individual's genomic information was part of the training data, as well as specific features of that individual's genomic information. Taking a different approach, individual anonymity can be kept through synthetic data generation, which uses the original data to create a new set of similar data to train the AI model. While this method

[97] Richard Annan et al., "Genomic Privacy and Security in the Era of Artificial Intelligence and Quantum Computing," Discover Computing 28, no. 1 (2025): 108, https://doi.org/10.1007/s10791-025-09627-w.
[98] Annan et al., "Genomic Privacy and Security in the Era of Artificial Intelligence and Quantum Computing."

does increase privacy when the new data is differentiated significantly from the original, there have been concerns with effectiveness of data utility and vice versa, resulting in this method being unable to fulfill both conditions.[99] Finally, there is a method called PPML-Omics that combines differential privacy and federated learning through a decentralized shuffling algorithm to ensure both significant utility of the data and anonymity of the patients. When applied to three deep-learning models to solve multi-omic tasks, it outperformed other privacy protection methods in efficiency and effectiveness while holding strong against cyber attacks.[100] Beyond its impressive results, PPML-Omics is especially notable because it is a combination of other data obfuscation methods, showcasing the constant innovations in privacy protection that help to address possible concerns over the safety of AI use in genomics. Through the aforementioned methods of differential privacy, federated learning, synthetic data generation, and PPML-Omics, there appears to be a general trend that AI models are becoming both more accurate and better at maintaining privacy.

Relevant Stakes for Data Privacy: Data Breach Possibilities

Patient data moves through many institutional layers across the United States, including Massachusetts. At each transition, the risk of a privacy breach increases. Data has shown that healthcare data breaches have significantly increased recently. As shown in Figure 1, 7,357 data breaches that affected over 500 people or more in healthcare were reported to OCR between 2009 and 2025. The graph highlights that the cases are increasing over time as well, which could

[99] Annan et al., "Genomic Privacy and Security in the Era of Artificial Intelligence and Quantum Computing."
[100] Juexiao Zhou et al., "PPML-Omics: A Privacy-Preserving Federated Machine Learning Method Protects Patients' Privacy in Omic Data," Science Advances 10, no. 5 (2024): eadh8601, https://doi.org/10.1126/sciadv.adh8601.

mirror the growing presence of AI in the field. Also, as technology continues to progress, the data becomes easier to trace back to the patient. Furthermore, the cost of genomic sequencing has decreased significantly, going from ~$300 million in 2000 to ~$700 million.[101] This has led to an increasing integration of genomics into care. Healthcare systems have increasingly been using electronic medical records (EMR) to store patient data. Genomic researchers progressively access this data as an easily accessible and inexpensive option for wide-ranging data. However, this involves many patients in involuntary medical research, often without their knowledge or consent.[102]

In fact, multiple hospitals in Massachusetts continue to intermittently have data breaches that greatly risk patient privacy. In 2019, Massachusetts General Hospital encountered unauthorized access to the computer applications containing health information of around 10,000 patients, according to the HIPAA journal.[103] The information – intended to be used for neurology research at MGH – included biomarkers and genetic information, in addition to basic clinical information such as dates of visits and tests, medical record number, diagnoses, treatment information.

While not common, data breaches have also affected other Boston area hospitals. In June 2024, Mass General Brigham fired three employees who granted patients' and policy holders' private information to an unauthorized, third party.[104] It is projected that this data breach specific

[101] Tim Dunn and Erdal Cosgun, "A Cloud-Based Pipeline for Analysis of FHIR and Long-Read Data," *Bioinformatics Advances* 3, no. 1 (2023): vbac095, https://doi.org/10.1093/bioadv/vbac095.
[102] Jennifer Kulynych and Henry T. Greely, "Clinical Genomics, Big Data, and Electronic Medical Records: Reconciling Patient Rights with Research When Privacy and Science Collide," *Journal of Law and the Biosciences* (2017): lsw061, https://doi.org/10.1093/jlb/lsw061.
[103] Alder, Steve. "Massachusetts General Hospital Data Breach Impacts 10,000 Patients. https://www.hipaajournal.com/massachusetts-general-hospital-data-breach-impacts-10000-patients/
[104] "Mass General Brigham Fires 3 Employees over 'Privacy Incident'." *NBC Boston*, June 28, 2024. https://www.nbcboston.com/news/local/mass-general-brigham-fires-3-employees-over-privacy-incident/3413879/.

to the Mass General Brigham Health Plan may have affected 3,659 individuals, in addition to the 655 individuals who were impacted by a breach in Mass General Brigham Incorporated.

Several months later in September 2024, Boston Children's Health Physicians faced a cyberattack and data breach by the Bianlian hacking group that accessed the personal and medical information of around 918,000 individuals, including both patients and employees according to the HIPAA journal.[105] This resulted in Boston Children's Health Physicians paying $5.15 million dollars to settle the lawsuit and remedy for the privacy damages of the data breach.

To begin the process of data collection, a healthcare provider will place an order to have a sample of biological matter sent to the laboratory. After testing, the data is captured by the laboratory information system, and then an electronic interface transfers it to an EMR.[106] However, clinical data can come in many forms, which would make exchange with other patient information difficult without a consistent standard. Currently, the top model is the 'Fast Healthcare Interoperability Resource' (FHIR®) format.[107] There are multiple steps to data processing in this model. First, patient data is needed. Often, tools like Synthea generate synthetic but realistic patient data in FHIR® format. Once stored in these platforms, the data can be compressed and downloaded. Afterwards, it can be uploaded to other online databases used

[105] Alder, Steve. "Boston Children's Health Physicians Pays $5.15M to Settle Data Breach Lawsuit." https://www.hipaajournal.com/boston-childrens-health-physicians-atsg-data-breach-settlement/.
[106] Abel N. Kho, Luke V. Rasmussen, John J. Connolly, Peggy L. Peissig, Justin Starren, Hakon Hakonarson, and M. Geoffrey Hayes, "Practical Challenges in Integrating Genomic Data into the Electronic Health Record," *Genetics in Medicine* 15, no. 10 (2013): 772–78, https://doi.org/10.1038/gim.2013.131.
[107] Tim Dunn and Erdal Cosgun, "A Cloud-Based Pipeline for Analysis of FHIR and Long-Read Data," *Bioinformatics Advances* 3, no. 1 (2023): vbac095, https://doi.org/10.109;generate3/bioadv/vbac095.

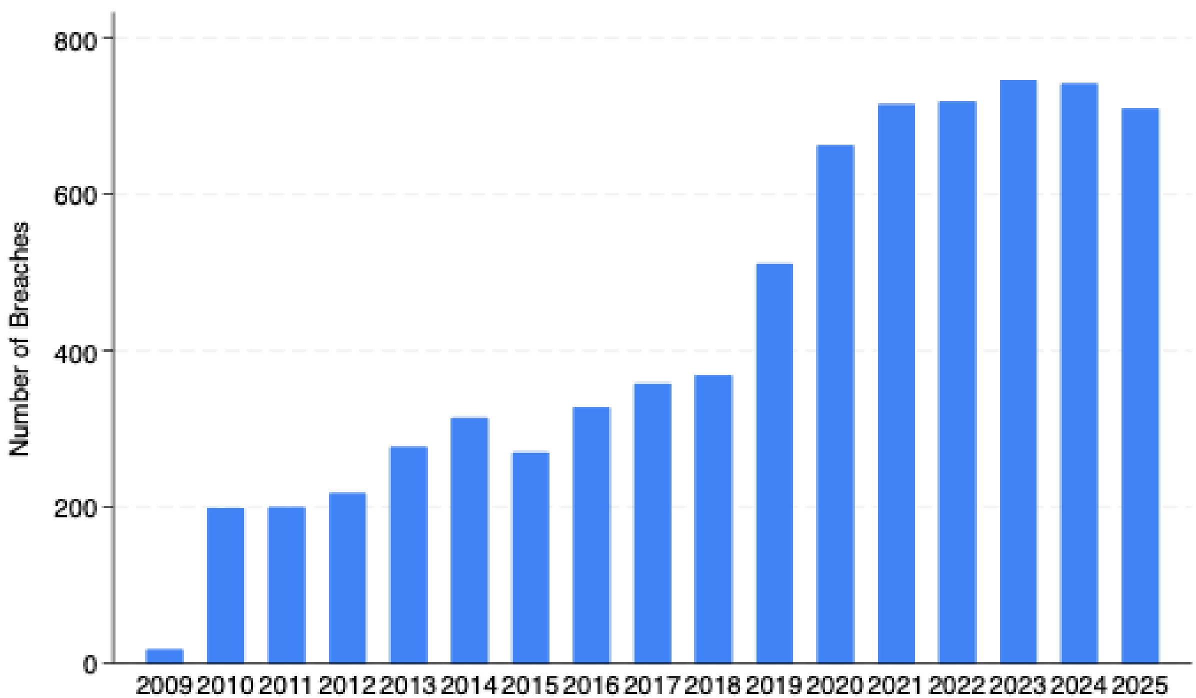
for medical storage or research.[108] At each step of data transmission, privacy risks increase. Copies of data are often stored in both the laboratory and hospital, and people who access the EMR may come across private genetic information.

Differential Increase in Data Breaches under LLM Utilization

Healthcare employees have been quick to adopt AI, with some sources reporting up to 66% adoption within the healthcare industry.[109] As more healthcare professionals adopt AI to assist their diagnoses and everyday tasks, more Americans' data is potentially exposed to unregulated, dangerous data processing. Most healthcare businesses have no formal agreements with LLMs to mitigate the potential risks associated with plugging patient PHI into an AI model. According to one source, "only about 23% of health systems report having Business Associate Agreements," which ensure compliance with HIPAA standards when dealing with third-party AI providers.[110] Because of this, LLMs that use inputs to further train their models run the risk of regurgitating protected patient information. AI models are also getting skilled at "re-identification," where the model can take several pieces of information (i.e., genomic data, phenotype, medical history) and piece together which patient is being referenced. This ability is not mitigated by HIPAA's "de-identification" clause, which mandates that 18 genetic markers be anonymized for all genomic datasets.

[108] Tim Dunn and Erdal Cosgun, "A Cloud-Based Pipeline for Analysis of FHIR and Long-Read Data," *Bioinformatics Advances* 3, no. 1 (2023): vbac095, https://doi.org/10.1093/bioadv/vbac095.
[109] Schoolcraft, Dave, Andrew C. Meltzer, Rohit Sangal, Aisha T. Terry, Katherine Robertson, Daniel Buckland, Sakib Motalib, Nicholas Genes, Rade Vukmir, and Tayab Waseem. "Health Insurance Portability and Accountability Act Liability in the Age of Generative Artificial Intelligence." JACEP Open 7, no. 2 (April 2026): 100317. https://doi.org/10.1016/j.acepjo.2025.100317.
[110] Schoolcraft, Dave, Andrew C. Meltzer, Rohit Sangal, Aisha T. Terry, Katherine Robertson, Daniel Buckland, Sakib Motalib, Nicholas Genes, Rade Vukmir, and Tayab Waseem. 2026. "Health Insurance Portability and Accountability Act Liability in the Age of Generative Artificial Intelligence." JACEP Open 7 (2): 100317. https://doi.org/10.1016/j.acepjo.2025.100317.

Artificial intelligence is also used by those looking to steal genetic or medical information. AI has been incredibly effective in lowering the cost of performing a cyberattack. It has also dramatically diversified the methods employed by hackers to gain access to PHI, likely causing the uptick in PHI leaks from 2019 to 2025. These effects of AI usage both within the medical sphere and by the bad actors looking to steal medical information from providers is incredibly concerning and should serve as a warning against the mass employment of AI in the healthcare industry.



Data Breach [Figure 1]: INSERT Description

Positively Reinforcing LLM Application Possibilities

Generative artificial intelligence is currently being implemented by the Center for Disease Control to assist in projects that address public health challenges, which has contributed to over $3.7 million estimated in labor costs saved to date and a 527% return on investment.[111] The CDC has used artificial intelligence to help formulate solutions for 55 public health challenges, and the CDC has additionally created an AI Accelerator, which aims to operationalize and scale the use of artificial intelligence to more complex public health problems. The current efforts to create health-based artificial intelligence models showcases that preliminary, broader applications of artificial intelligence into health sectors are working. When specifically considering the Massachusetts area, "Scientists at the Broad Institute and Mass General Brigham have built a generative AI model [called DNA-Diffusion] that creates short DNA segments that can control gene activity in specific cells." [112] Through the use of this model in conjunction with existing genetic-based treatments, the researchers found that it was possible to target and activate specific cancer defense genes, which opens up new possibilities in cancer treatment. The same process of targeting genes can apply to many other treatment methods, and this entire process of applying artificial intelligence to genomics is thus shown to be already underway within Massachusetts.

[111] CDC. "CDC's Vision for Using Artificial Intelligence in Public Health." Center for Disease Control, September 18, 2025. https://www.cdc.gov/data-modernization/php/ai/cdcs-vision-for-use-of-artificial-intelligence-in-public-health.html.

[112] Colarossi, Jessica. "AI Generates Short DNA Sequences That Show Promise for Gene Therapies." Broad Institute, January 23, 2026. https://www.broadinstitute.org/news/ai-generates-short-dna-sequences-show-promise-gene-therapies.

Representation Bias Concerns

There are growing concerns about data that reflect systemic challenges and the structures used in AI systems for training and testing. Concerns stem from the over-representation of European ancestry, since it may reflect only a small portion of global DNA, and from the influence of societal biases in genomic data. [113][114] This creates a representation bias, where minority populations are inadequately represented in genetic screening databases, creating significant healthcare disparities as AI systems may not accurately predict disease risks or treatment responses for underrepresented groups.[115] Imbalances in representation within genomic datasets can cause AI-based prediction and decision-support systems to perform unevenly, with smaller patient subgroups receiving less reliable outputs than more represented populations.[116] Data quality problems, including misclassification and measurement error, can compound these disparities when inaccuracies are patterned by setting, workflow, or patient characteristics rather

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[113] Hanna, Matthew G., Liron Pantanowitz, Brian Jackson, Octavia Palmer, Shyam Visweswaran, Joshua Pantanowitz, Mustafa Deebajah, and Hooman H. Rashidi. 2025. "Ethical and Bias Considerations in Artificial Intelligence/Machine Learning." *Modern Pathology* 38 (3): 100686. https://doi.org/10.1016/j.modpat.2024.100686.

[114] Mckibbin, Popejoy, Shabani, *Reconciling Diversity in Helth and Genomic Data, 1-2*

[115] Hanna, et. al., *Ethical and Bias Considerations, 1-4*

[116] Hanna, et. al., *Ethical and Bias Considerations, 4*

than randomly distributed.[117] Patients from lower-resource contexts who receive care may be particularly exposed when documentation standards and data entry practices differ across sites.

Second, Massachusetts must address bias in genomic AI model training through mandatory evaluation and transparency requirements. Genomic datasets historically overrepresent individuals of European ancestry. In a genomic sequencing study within The Cancer Genome Atlas analyzing race and ethnic disparities, Spratt et. al found that over 75 percent of patients identified as white, leading to the probable conclusion that this overrepresentation and lack of ethnic diversity is related to cancer development.[118] Furthermore, a landmark analysis by Martin et al. found that over 80 percent of participants in genome-wide association studies were of European descent. This raised concerns that polygenic risk scores derived from these datasets perform worse in non-European populations.[119] Both studies warn that clinical use of these scores and conclusions could exacerbate health disparities. If Massachusetts permits widespread adoption of genomic AI tools without addressing dataset imbalance, predictive tools may systematically underperform for non-white patients.

Since 2020, over 100 rural hospitals have shuttered their doors[11], drastically outpacing urban hospital closures[120]. This is the most dramatic indication of the destructive divide between

[117] Gianfrancesco, Milena A., Suzanne Tamang, Jinoos Yazdany, and Gabriela Schmajuk. 2018. "Potential Biases in Machine Learning Algorithms Using Electronic Health Record Data." JAMA Internal Medicine 178 (11): 1544. https://doi.org/10.1001/jamainternmed.2018.3763.

[118] Spratt, Daniel E, et al. "Racial/Ethnic Disparities in Genomic Sequencing." *JAMA Oncology*, U.S. National Library of Medicine, 1 Aug. 2016, pmc.ncbi.nlm.nih.gov/articles/PMC5123755/. Accessed 24 Feb. 2026.

[119] Alicia R. Martin et al., "Clinical Use of Current Polygenic Risk Scores May Exacerbate Health Disparities," *Nature Genetics* 51 (2019): 584–591.

[120] Office, Accountability. 2025. "Urban Hospitals: Factors Contributing to Selected Hospital Closures and Related Changes in Available Health Care Services." Gao.gov. 2025. https://www.gao.gov/products/gao-25-106473.

rural and urban healthcare availability, but it does not end there. In the hospitals that are not closed, rural hospital staff is fewer and farther in between[121]. They are also dramatically underpaid compared to their urban hospital counterparts[122]. Even the hospitals that are properly staffed are almost certainly lacking the critical IT and tech experts necessary to maintain the high-tech infrastructure of large urban hospitals[123]. Even if these fully-staffed, rural hospitals do have IT and data-science professionals on site, there is a possibility that they will not even have access to broadband or wifi[124]. While there is no registry that pinpoints the exact number of hospitals or clinics without high-speed wifi access, Pew Charitable Trust estimates that as many as 60% of healthcare facilities outside of metropolitan areas lack broadband access[125]. In addition, the FCC estimates that as many as 20 million Americans lack broadband access in their communities or homes[126]. These disparities make it harder to run a modern, connected healthcare facility, and in the digital age, exacerbate the number of patients who will choose an urban facility over a more local rural facility, which can lead to more closures. Technology has deeply ingrained itself into the healthcare landscape, and rural communities often face insurmountable obstacles to modernization[127]. This modernization gap can quickly force hospitals to close –

[121] "NRHA's Rural Health Voices Blog | National Rural Health Association - NRHA | NRHA." 2021. National Rural Health. 2021.
https://www.ruralhealth.us/blogs/2025/07/recruitment-challenges,-solutions,-and-outlooks-for-the-rural-doc-shortage
[122] Sablik, Tim. 2022. "The Rural Nursing Shortage." Richmondfed.org. Econ Focus. February 28, 2022.
https://www.richmondfed.org/publications/research/econ_focus/2022/q1_feature_1#:~:text=But%20as%20COVID%2D19%20caseloads,scaled%20back%20during%20the%20pandemic..
[123] "Health Information Technology Value in Rural Hospitals | Digital Healthcare Research." 2023. Ahrq.gov. 2023.
https://digital.ahrq.gov/ahrq-funded-projects/health-information-technology-value-rural-hospitals.
[124] Bervell, Joel. 2025. "In Rural America, a Weak Signal Can Mean Worse Health." The Commonwealth Fund Website, May. https://doi.org/10.26099/0zyb-fh21.
[125] Winslow, Joyce. 2019. "America's Digital Divide." Pew.org. The Pew Charitable Trusts. July 26, 2019.
https://www.pew.org/en/trust/archive/summer-2019/americas-digital-divide.
[126] "Seventh Broadband Progress Report." 2025. Fcc.gov. 2025.
https://www.fcc.gov/reports-research/reports/broadband-progress-reports/seventh-broadband-progress-report.
[127] Bennett, Kevin, Celli Horstman, Corinne Lewis, and Anthony Shih. 2026. "Why Rural Hospitals Are Facing a Funding Crisis — and How It Could Get Worse." The Commonwealth Fund Website, February.
https://doi.org/10.26099/bywa-x418.

explaining the difference between urban and rural hospital closures – and is being sped up by the advent of AI in healthcare[19].

Upon preliminary analysis, it appears as if the urban-rural healthcare divide could be dramatically reduced by supporting rural healthcare facilities to adopt emerging AI technologies. Without adopting such approaches, it is clear the rural hospitals will face an uphill-battle to stay in operation. Without adequate access to local healthcare facilities, rural populations – who already face higher rates of many preventable diseases – may find themselves living in healthcare deserts. This destruction of rural healthcare would have other latent effects as well. Large hospitals, sensing an advantage in early AI adoption, will increase their investments, leading to an increasing reliance on AI for every task. The effects of such widespread reliance is speculated on, but it could lead to catastrophic results if not properly regulated. The United States could see a transportation renaissance as the need for faster commutes to urban epicenters increases. On the other hand, rural communities themselves, without proper healthcare infrastructure, could die off as a way of life if living removed from hospitals and convenient urban infrastructure becomes impossible. This is, of course, a worst-case-scenario perspectivetake on the future of healthcare, but all of this is distinctly not impossible without action to level the technological playing field. Access. While this serves merely as an introduction to the research that will occur in the weeks to come, it seems increasingly unlikely that any related policy recommendation would not include a way to bridge the gap between urban and rural access to emerging healthcare technology. Failing to do so could be catastrophic.

Furthermore, there is a proven lack of diversity in genomic data, which creates an inaccurate representation of global genomic diversity. This potentially lies the foundation for inequitable predictive AI systems. Manuel Corpus et al. bring up in their article "Bridging Genomics' Greatest Challenge: The Diversity Gap," data from the GWAS Diversity Monitor showing that "The total proportion of representation from any population other than Asian (3.96%) or European (94.48%) is <1%."[128] This inequality means that predictive models utilizing this data may perform less well for underrepresented groups, increasing the likelihood of unequal treatment recommendations. Although this is not a direct legal violation, employing inequitable AI tools raises concerns related to anti-discriminatory or civil rights frameworks.

Challenges to Differential Privacy Methodologies

Yet one of the most fundamental challenges that keeps resurfacing is that no two individuals share an identical genome; population level accuracy does not guarantee individual level accuracy. But with the exception of identical twins, each person's DNA sequence is unique which means a DNA sample can't ever be truly anonymized, but needs to be used for ensuring a higher individual level accuracy. Surprisingly, a study from 2013 showed that participants can be r-identified using genomic data from a dataset prairie with genealogical databases and public records. Thus differential privacy is increasingly being used for protecting individual-level data in AI model training. It works by adding random noise to analysis results. It stems off of the idea that more noise leads to better privacy but degrades the utility of the result (it is a tradeoff). In

[128] Manuel Corpas et al., "Bridging Genomics' Greatest Challenge: The Diversity Gap," *Cell Genomics* 5, no. 1 (January 8, 2025): 100724, https://doi.org/10.1016/j.xgen.2024.100724

the AI training context, the DPSGD (differentially private stochastic gradient descent) serves as the key mechanism, essentially where the noise is injected during the model training process. Across modalities like DP-SGD, there is a maintenance of clinically acceptable performance, but strict privacy with an epsilon value of around 1 leads to substantial accuracy loss so there is an extent to which this stochastic noise can be utilized. It is also worth mentioning that differential privacy hurts most where datasets are the smallest, particularly that of rare genetic disorders since it is simply the nature of how many patients have the condition.

Legislative Contextualization and Assessment of Threat Imminence

Federal Context [needs to be packaged into complete sentences]

1. **Privacy Act of 1974**[129]
    a. Federal law that regulates how executive branch agencies collect, maintain, use, and disclose personally identifiable information of US citizens and permanent residents. Grants individuals the right to access, review, and amend their records. Restricts unauthorized disclosure.
    b. Implications:
        i. Established one of the first federal frameworks for protecting personal data held by government agencies, which shaped how federal health and research databases handle sensitive information.

[129] "Privacy Act of 1974, 5 U.S.C. § 552a | Bureau of Justice Assistance," Bureau of Justice Assistance, n.d., https://bja.ojp.gov/program/it/privacy-civil-liberties/authorities/statutes/1279.

ii. Explicitly created individual rights over personal records, like the ability to access and correct data, which later influenced other health and genomic data privacy policies.

iii. Set an early precedent for limiting unauthorized disclosure of sensitive information, an idea that later privacy laws, like HIPAA, expanded to medical and genetic data.

2. **Belmont Report & Human Subjects Research Framework**[130]
   a. By the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, which established the core ethical framework for research involving human subjects in the US, and contains 3 fundamental principles:
      i. **Respect for Persons**: voluntary and informed participation
      ii. **Beneficence**: researchers protect subjects from harm by maximizing potential benefits and minimizing risks
      iii. **Justice**: equitable distribution of research burdens and benefits. No group to be unfairly selected/exploited
   b. Implications:
      i. Became the ethical foundation for all human-subject research in the United States, including biomedical and genomic research.[131]

[130] "The Belmont Report - UW Research." 2024. UW Research. January 26, 2024. https://www.washington.edu/research/hsd/guidance/ethical-principles/belmont/.
[131] Protections, Office for Human Research. 2024. "The Belmont Report." HHS.Gov. October 17, 2024. https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/index.html.

ii. Established the requirement for informed consent, where individuals must understand and voluntarily agree to participate in research involving their personal or genomic information.

iii. Introduced ethical standards that protect vulnerable populations, ensuring equitable participation and protection from exploitation in medical research.

iv. Led directly to the creation of federal regulations governing research ethics, including Federal Policy for the Protection of Human Subjects.

3. **The Common Rule (Federal Policy for the Protection of Human Subjects) 1991**[132]

   a. A federal ethical regulation that protects human subjects in research supported/conducted by federal departments. Requires Institutional Review Board review, approval, and informed consent.

   b. Implications:

      i. Operationalized the Belmont Report by creating federal regulations governing human-subject research across government-funded institutions.

      ii. Requires Institutional Review Board oversight, ensuring that research involving patient or genomic data undergoes ethical review before it can proceed.

      iii. Strengthened consent, privacy protections, and risk assessment in medical care and research, which are particularly important for studies involving large-scale genetic datasets.

[132] Protections, Office for Human Research. 2024. "The Belmont Report." HHS.Gov. October 17, 2024. https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/index.html.

Overview of Seminal Regulatory Frameworks

In the era of big data genomics, transparency functions as the primary mechanism through which privacy is operationalized and trust is maintained. Informed consent processes, disclosures of financial relationships, and institutional data governance policies all determine whether patients perceive their information is being used ethically.9 In Massachusetts, the status quo reflects a layered governance model where privacy is shaped by federal law, state statutes, and professional norms. However, failures in transparency often arise not from an absence of regulation, but from breakdowns in institutional culture, such as unauthorized internal access or secondary data use without clear communication.3 Effective transparency is critical because public willingness to share data is highly dependent on the purpose of the disclosure. Research indicates that the specific use of data (e.g. academic research vs. commercial marketing) is a significantly more important factor to patients than the identity of the user or even the sensitivity of the data itself.8 When transparency mechanisms fail to clearly articulate these purposes, patients may withhold sensitive information, which ultimately diminishes the quality of clinical care and the strength of the medical evidence base.3,8 Thus, transparency is not merely an administrative hurdle but a structural bridge between legal protections and the actual patient experience.9

At the federal level, the Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule and the Genetic Information Nondiscrimination Act (GINA) provide the foundational legal boundaries for protecting health information.5,7 HIPAA establishes a governance framework that balances individual privacy with the "flow of health information" necessary for high-quality care, primarily through its "Notice of Privacy Practices."5 However, HIPAA allows significant flexibility, permitting data sharing without explicit consent for treatment, payment, and healthcare operations.5 This creates a "mirage" of privacy, as genomic data is "inherently and permanently identifiable," making traditional anonymization such as stripping names or addresses insufficient for true protection.4,7 The uniqueness of the human genome means that re-identification remains a persistent risk even in "de-identified" datasets.4 For example, research has demonstrated that surnames can be recovered from personal genomes by profiling short tandem repeats on the Y chromosome and querying publicly accessible genetic genealogy databases.4 Furthermore, because genetic information is shared among biological relatives, one individual's disclosure can reveal the disease predispositions of family members who never consented to testing.6 This "identifiability paradox" suggests that traditional one-time consent models are inadequate for managing long-term biobanking and the integration of genomic data into large AI-driven datasets.6,7

Shortcomings of HIPAA and Existing Frameworks of Governance

HIPAA was created with three main clauses. The privacy clause generally guarantees that Private Health Information (PHI) belongs to the patient alone, and that distribution of said material is generally only permissible with the patient’s consent.[133] The second clause, the security clause, exhibits what forms of patient information are considered PHI and how they must be properly protected (or transferred, in the case of electronic Private Health Information (ePHI)).[134] The final clause, the breach notification rule, says that covered entities (healthcare

[133] “SUMMARY of the HIPAA PRIVACY RULE HIPAA Compliance Assistance.” 2003. https://www.hhs.gov/sites/default/files/privacysummary.pdf.
[134] Office. “Summary of the HIPAA Security Rule.” HHS.gov, August 13, 2025. https://www.hhs.gov/hipaa/for-professionals/security/laws-regulations/index.html.

providers and health insurance companies that fall under the purview of HIPAA regulations) must report all breaches of PHI to those affected.[135]

HIPAA has only been amended five times since it was signed into law in 1996, despite massively increasing occasions of PHI breaches and the advent of AI-assisted healthcare tools.[136+1] Failure to amend existing laws leaves patients vulnerable to increasing PHI breaches and the devastating consequences that can come with them. A lack of AI privacy regulations has left millions of Americans vulnerable to dangerous PHI leaks, and HIPAA must be amended to reflect the dangers that AI poses to patient data.

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First, Massachusetts should adopt enhanced protections for patient genomic data that go beyond current federal guidelines. While the federal Health Insurance Portability and Accountability Act (HIPPA) provides baseline health protections—such as access to records, correction requests to health information, and providing written consent for using and/or disclosing health information—it fails to fully address emerging AI-specific risks and the

[135] Office. "Breach Notification Rule." HHS.gov, August 13, 2025. https://www.hhs.gov/hipaa/for-professionals/breach-notification/index.html.
[136] Khaja, Abdullah, Aftab Tariq, Michidmaa Arikhad, and Tamoor Sadiq. 2025. "The HIPAA Singularity: Reconciling Artificial Intelligence, Cybersecurity, and Patient Rights in the U.S. Healthcare Legal Framework." https://doi.org/10.56472/25835238/IRJEMS-V4I9P105.
[137] Khaja, Abdullah, Aftab Tariq, Michidmaa Arikhad, and Tamoor Sadiq. 2025. "The HIPAA Singularity: Reconciling Artificial Intelligence, Cybersecurity, and Patient Rights in the U.S. Healthcare Legal Framework." https://doi.org/10.56472/25835238/IRJEMS-V4I9P105.

commercial uses of genomic data.[138] Genomic information is uniquely identifiable and *cannot* be meaningfully anonymized in the same way as other health data can. Erlich and Narayanan have demonstrated that supposedly de-identified genetic data can often be re-identified by linking it with publicly available information, undermining assumptions of anonymity.[139]

Law 70G and Relevant Extensions

Currently, Massachusetts has various regulations and legislation already in place to safeguard the privacy and autonomy of individual patients in their genomic data. Most notably, Massachusetts General Law Part 1 Title XVI Chapter 111 Section 70G defines individual genetic information and reports as completely private and never public.[140] This section details what falls under genetic information used for clinical purposes and the specific measures that hospitals and researchers have to follow to prevent patient privacy abuse. Its emphasis is particularly on ensuring prior informed written consent before both genetic testing and any disclosing of data. Violation of these measures entails penalties under unfair practices in Massachusetts' Consumer Protection Law.[141] While being the only section in MGL that explicitly addresses genomic data, it leaves minimal room for exceptional circumstances and overall demonstrates thorough

[138] U.S. Department of Health and Human Services. "The HIPAA Privacy Rule." *HHS.Gov*, 13 Aug. 2025, www.hhs.gov/hipaa/for-professionals/privacy/index.html#:~:text=The%20HIPAA%20Privacy%20Rule%20establishes,160%2C%20162%2C%20and%20164. Accessed 24 Feb. 2026.
[139] Yaniv Erlich and Arvind Narayanan, "Routes for Breaching and Protecting Genetic Privacy," Nature Reviews Genetics 15, no. 6 (2014): 409–421.
[140] "General Law - Part I, Title XVI, Chapter 111, Section 70G." Accessed February 24, 2026. https://malegislature.gov/Laws/GeneralLaws/PartI/TitleXVI/Chapter111/Section70G.
[141] "General Law - Part I, Title XV, Chapter 93A, Section 2." Accessed March 3, 2026. https://malegislature.gov/Laws/GeneralLaws/PartI/TitleXV/Chapter93a/Section2.

guidelines on healthcare providers. However, the detail of regulations somewhat appears to be more exclusive to specifically genomic data because Massachusetts law on patient privacy of general medical data is considered less stringent than the regulations listed by Health Insurance Portability and Accountability Act (HIPAA), compared to other state and federal laws.[142]

As many hospitals in Massachusetts utilize genomic data and implement AI technology initiatives, the concern for patient privacy is an especially relevant issue. Massachusetts exhibits various regulations and legislation in place that addresses its possible implications. For example, MGL Part 1 Title XVI Chapter 111 Section 70G outlines the protection of individual genetic information and reports protected as private information, as well as prior written consent for genetic testing.[143] This section comprehensively defines and details what falls under genetic information that is used for clinical purposes and what specific measures hospitals and researchers have to strictly follow to prevent the abuse of patient privacy. In addition, a bill (Bill S.46) has been recently passed in 2025 that amends Section 12 of Chapter 176O of the General Laws by adding a subsection specifically regarding ethical and practical guidelines on the use of artificial intelligence and other software tools in healthcare decision-making.[144]

Third, Massachusetts should clarify legal responsibility in cases involving AI-assisted genomic decision-making. Current malpractice law holds physicians to a reasonable standard of

[142] "Guide on the disclosure of confidential information: Health care information." Accessed March 3, 2026. https://www.mass.gov/info-details/guide-on-the-disclosure-of-confidential-information-health-care-information.

[143] "General Law - Part I, Title XVI, Chapter 111, Section 70G." Accessed February 24, 2026. https://malegislature.gov/Laws/GeneralLaws/PartI/TitleXVI/Chapter111/Section70G.

[144] "Bill S.46." Accessed February 24, 2026. https://malegislature.gov/Bills/194/S46.

care, a legal term that "refers to the degree of care a prudent and reasonable person would exercise over" specific circumstances.[145] But, the integration of AI complicates this standard. If a physician relies on an AI-generated genomic risk assessment that later turns out to be inaccurate due to a model defect, it raises the question of who or what to hold accountable. Mello et al. argues that without clear policy guidance, liability uncertainty may unfairly insulate developers from accountability.[146] Because of this, Massachusetts should enact legislation explicitly defining genomic AI systems as clinical decision-support tools rather than autonomous decision-makers. This method preserves physician responsibility for final clinical judgments. However, the statute should also recognize shared liability where harm results from algorithmic defects, specifically if developers failed to disclose known performance limitations. Additionally, Massachusetts should require a baseline level of explainability for AI systems used in high-stakes genomic decision-making. Requiring interpretable outputs and documentation of model reasoning would enhance transparency and strengthen patient trust.

Case Studies Concerning Data Privacy Considerations

Example 1: 23andMe Data Breach Investigation

Re‑identification research has demonstrated that, given demographic attributes, 99.98% of Americans can be correctly re‑identified in a supposedly anonymized dataset with only 15

[145] Vanderpool, Donna. "The Standard of Care." *Innovations in Clinical Neuroscience*, U.S. National Library of Medicine, 2021, pmc.ncbi.nlm.nih.gov/articles/PMC8667701/. Accessed 24 Feb. 2026.
[146] Michelle M. Mello et al., "Shifting Liability in the Era of Artificial Intelligence in Health Care," *New England Journal of Medicine* 382 (2020): 2041–2043.

attributes[147]. For predictive AI models, this means "de-identified" training data is often effectively identifiable, especially when combined with genomic data. From major data breaches from the London hospital and 23andMe, it remains imperative to protect and anonymize genomic related data to prevent any further reidentification.

In late 2023, popular DNA testing company 23andMe disclosed a breach that compromised the personal and genetic data of millions of users. A hacker calling themselves "Golem" began leaking datasets allegedly stolen from 23andMe accounts categorized by ethnic groups. Initially Ashkenazi Jewish and Chinese users had their data leaked, indicating a possible intent to target specific populations, and the leak eventually expanded to include over 6.9 million profiles. Roughly 14,000 accounts were directly accessed through reused passwords, but each account's participation in the "DNA Relatives" feature allowed the attacker to scrape the profile data of millions of related users[148]. Due to the nature of genomic data, leaked data does not only impact one user but their entire family tree, highlighting the importance of security for genomic data.

Example 2: Royal Free Hospital and Google DeepMind

The UK's data protection authority has found that in 2015, London's Royal Free hospital unlawfully gave the health records of 1.6 million patients to Google's cutting-edge artificial intelligence unit DeepMind[149]. Regulators later concluded that this large-scale data sharing was a

[147] Rocher, L., Hendrickx, J.M. & de Montjoye, YA. Estimating the success of re-identifications in incomplete datasets using generative models. *Nat Commun* 10, 3069 (2019). https://doi.org/10.1038/s41467-019-10933-3.
[148] Demyd Maiornykov. "The 23andMe Breach: Anatomy, Impact, and Lessons for Genomic Security (Deep Dive)." *Sekurno*, 7 Apr. 2025, www.sekurno.com/post/the-23andme-breach-anatomy-impact-and-lessons-for-genomic-security-deep-dive.
[149] Wong, Joon Ian, and Joshua Wong. "DeepMind Held 1.6 Million People's Health Records Unlawfully, but No One's Getting Fined for It." *Quartz*, 4 July 2017,

clear breach of data protection laws: patients were not informed that their identifiable health records would be used to develop AI systems[150]. This is another clear example of genomic and health related data being unlawfully used to train AI models. Further, this adds to potential bias concerns being baked into AI models, as the London hospital largely had medical records of people from European descent, overrepresenting a particular demographic in AI models.

Following the data break from the London hospital, the ICO required Royal Free to implement stronger transparency, conduct a third-party audit, and revise its data-sharing agreement with DeepMind to ensure lawful processing and clear information for patients[151]. However, the damage has been done: millions of users had their data leaked across the internet and to third party actors, leaving lasting impacts including re-identification.

Example 3: Mass General BioBank

Patients in the Mass General Brigham hospital system have the choice of opting in to submitting their medical information to the Mass General Brigham Biobank. The Biobank links test data, pharmacological prescriptions, and patient outcomes for use in health research.[152] In particular, over 65,000 patients have volunteered their genomic data for the use of health research.[153] Researchers use this genomic information in combination with electronic health

qz.com/1020473/deepmind-held-1-6-million-peoples-health-records-unlawfully-but-no-ones-getting-fined-for-it. Accessed 3 Mar. 2026.
[150] Burgess, Matt. "NHS DeepMind Deal Broke Data Protection Law, Regulator Rules." *Wired*, 3 July 2017, www.wired.com/story/google-deepmind-nhs-royal-free-ico-ruling/.
[151] "Google DeepMind and Class Action Lawsuit." *Ico.org.uk*, ICO, 15 Nov. 2024, ico.org.uk/for-the-public/ico-40/google-deepmind-and-class-action-lawsuit/. Accessed 3 Mar. 2026.
[152] "Biobank | Mass General Brigham." Accessed February 22, 2026. https://www.massgeneralbrigham.org/en/research-and-innovation/participate-in-research/biobank
[153] "How the Biobank Is Driving Medical Discovery | Mass General Brigham." Accessed February 22, 2026. https://www.massgeneralbrigham.org/en/about/newsroom/articles/biobank-driving-medical-discovery

record data to investigate how genetic variation influences disease risk, treatment response, and long-term health outcomes. Because the Biobank connects DNA samples with detailed clinical histories, scientists are able to study patterns across large populations rather than isolated cases. This allows for the identification of genetic markers associated with common conditions such as cardiovascular disease, diabetes, cancer, and psychiatric disorders.

In addition to supporting broad research efforts, the Biobank also plays a role in translating discoveries back into patient care. When research uncovers genetic findings that are considered medically actionable, participants may be contacted and offered confirmatory clinical testing and genetic counseling. In these cases, information that began as research data can ultimately inform individualized prevention strategies, enhanced screening protocols, or targeted treatments.

At the forefront of advancing healthcare in the United States, Massachusetts has implemented multiple AI-based technologies for genomic data analysis and interpretation. In particular, Mass General Brigham has launched many novel initiatives to incorporate AI and genomic data analysis into both patient care and clinical research. One most commonly known program is BioBank, MGB's major research program that investigates the manner in which genomic factors affect health and disease by recruiting patients and collecting their blood sample data.[154] As of 2024, BioBank recruited over 150,000 patients to distribute around 250,000 samples and specifically collected genomic data from 65,000 participants that were used for

[154] "Biobank Overview | Mass General Brigham." Accessed February 24, 2026. https://www.massgeneralbrigham.org/en/research-and-innovation/participate-in-research/biobank/overview.

more than 600 unique studies.[155] BioBank has supported the analysis of several pressing medical questions, which include the relationship between biomarkers and risk of developing cancer, as well as the effect of genes on steroid medication response, surgery confusion, and risk for preeclampsia. Storing 1.3 million biospecimens, BioBank has significantly enhanced efficiency of collecting sample data from patients that is necessary for scientific and genomic research, accelerating efforts to address key medical conundrums today. Meanwhile, BioBank also lays out regulations for protecting patient privacy, adhering to federal laws such as the Health Insurance Portability and Accountability Act (HIPAA) and, in particular, Genetics Information Non-Discrimination Act (GINA) – a federal law that protects abuse of genetic information against individuals applying for employment or health insurance.[156] By also de-identifying samples or working with the Institutional Review Board (IRB), BioBank protects the privacy and confidentiality of individual information when used in research.

In more direct clinical settings, Mass General Brigham has recently pioneered in December 2025 a spinout AI technology company that developed a generative AI software to efficiently screen and enroll patients for clinical trials.[157] This software tool called RECTIFIER (RAG-Enabled Clinical Trial Infrastructure for Inclusion Exclusion Review) makes use of the Retrieval-Augmented Generation (RAG) AI application. This software reviews electronic health

155 "How the Biobank Is Driving Medical Discovery | Mass General Brigham." Accessed February 24, 2026.
https://www.massgeneralbrigham.org/en/about/newsroom/articles/biobank-driving-medical-discovery.
156 "Protecting Your Privacy | Mass General Brigham." Accessed February 24, 2026.
https://www.massgeneralbrigham.org/en/research-and-innovation/participate-in-research/biobank/protecting-your-privacy.

157 "Mass General Brigham Announces New AI Company to Accelerate Clinical Trial Screening and Patient Recruitment | Mass General Brigham." Accessed February 24, 2026.
https://www.massgeneralbrigham.org/en/about/newsroom/press-releases/aiwithcare-mass-general-brigham-spinout-new-company.

records of patients to understand their eligibility for clinical trials in key health problems such as heart failure, sickle cell disease, and amyotrophic lateral sclerosis, significantly improving the efficiency of patient enrollment twofold. While this AI-based technology is not specifically geared toward patient data, it exemplifies Mass General Brigham's efforts to implement AI in patient recruitment for clinical research, which may include research projects that BioBank supports with its collected blood sample data from numerous patients.

Example 4: OncoNPC

The Dana-Farber Cancer Institute showcases another example of predictive AI-based technology that specifically uses gene sequencing data to understand cancer treatment. Called OncoNPC (Oncology NGS-based Primary cancer type Classifier), this predictive technology assists doctors in identifying the tumor source and selecting relevant treatments that improve patients' survival.[158] OncoNPC's machine learning model is based on retrospective medical and genomic records—specifically, tumor genetic sequencing data and clinical information—of 36,445 patients who have been identified with primary tumors from three cancer centers including Dana-Farber. This data also helps researchers and doctors using OncoNPC to identify genetic factors that influence the model's prediction. With an 80% accurate prediction of the tumor source and high confidence predictions in multiple applications and research studies, the technology has led to real improvements in outcomes such as more patients being matched to their target treatment and longer survival.

[158] "Dana-Farber AI-Model Predicts Primary Source of Cancer Using Gene Sequencing Data - Dana-Farber." August 15, 2023. http://physicianresources.dana-farber.org/news/dana-farber-ai-model-predicts-primary-source-of-cancer-using-gene-sequencing-data.

State-wide Comparative Frameworks and Assessments

Regulatory Framing and Recommendations

Limitations to Federated Learning Approaches

It is already well known that patient genomics is being used to train AI models, particularly data that has been considered de-identified, while also linking clinical phenotypes to enable supervised learning. Going beyond centralized datasets that train predictors or other ML models, though, there is also much interest in today's community in federated learning[159] for genomics, in which hospitals or research institutes train a shared model locally and share only encrypted model updates as opposed to raw genomic data. But even with a broader attempt to alleviate privacy risks, recent investigations into federated learning in genomics shows that even when raw data stay local, model updates can still end up leaking sensitive information through membership and inference attacks, which are quite common in today's age.[160]

[159] Sk. Tanzir Mehedi, et al. "A Privacy-Preserving Dependable Deep Federated Learning Model for Identifying New Infections from Genome Sequences." Scientific Reports, vol. 15, no. 1,
[160] Kokje, Yashashree. Privacy Preserving Framework for Federated Learning in Genomics. 2020.
Pathade, Chetan, and Shubham Patil. "Securing Genomic Data against Inference Attacks in Federated Learning Environments." ArXiv.org, 2025, arxiv.org/abs/2505.07188. Accessed 24 Feb. 2026.

II: Policy Recommendations

Amend *Bill S.2619:* An Act establishing the Massachusetts data privacy act

To address the issue of insurance companies potentially using AI systems to generate genetic risk scores by bypassing genetic testing limitations, I propose an additional amendment to *Bill S.2619:* An Act establishing the Massachusetts data privacy act, which is currently being considered in the Massachusetts House of Representatives after passing in the Massachusetts Senate, to ban insurance companies from considering genetic information when issuing insurance policies to residents of Massachusetts.[161] *Bill S.2619* currently would limit collection, sharing, and sale of sensitive data such as health records, genetic/genomic data, biometric identifiers (e.g., DNA, facial data), precise location data, and data about minors through explicit opt-in consent and requiring specified purposes, as well as allowing residents of Massachusetts to access, correct, delete, and transfer their data.[162] However, once insurance companies procure consent to access information such as health records, *Bill S.2619* does not currently limit how insurance companies can use that information to perform genetic discrimination. As outlined prior, AI has shown capabilities to infer a patient's genetic risk without requiring genetic testing, which would bypass many of the current restrictions on insurance companies that are focused specifically on genetic testing.[163] A complete ban on the use of genetic information by insurance companies in their coverage plans is also not entirely new in Massachusetts, since for health

[161] Senate Committee on Ways and Means, "Bill S.2619," The 194th General Court of the Commonwealth of Massachusetts, Accessed March 22, 2026, https://malegislature.gov/Bills/194/S2619.
[162] The 194th General Court of the Commonwealth of Massachusetts, "Bill S.2619."
[163] "General Law - Part I, Title XVI, Chapter 111, Section 70G."

insurance, under the Massachusetts Genetic Privacy Act (Acts 200 Chapter 254), genetic tests are prohibited from being used as a requirement for health insurance and genetic information is already not allowed to be used as the basis for discriminatory health insurance plans.[164,165] The issue is that life, disability, and long-term care insurance is covered under Massachusetts General Law Part I, Title XXII, Chapter 175, Section 120E, which allows genetic discrimination in insurance policy decisions as long as the discrimination is not deemed to be unfair discrimination, defined as the altering of insurance policy that is not grounded in reasonable belief of an individual's increased risk of mortality.[166] Because an AI generated risk score would provide ample evidence of an individual's increased risk of mortality due to genetic diseases, there are essentially no current restrictions on genetic discrimination by life, disability, and long-term care insurance companies. By including an additional provision in *Bill S.2619* to expand the prohibition on the use of genetic information insurance decisions from just health insurance companies to instead cover all insurance companies within Massachusetts, residents of Massachusetts will not have to worry that the new, promising developments of AI applications in genomics will be used to price them out of insurance or to change their coverage in any way. The reason to include the expansion of restrictions on genetic insurance discrimination as a part of *Bill S.2619* is because it would align with the goal of the bill to protect the privacy rights of Massachusetts residents, and *Bill S.2619* has already passed the Massachusetts Senate and is currently being discussed for amendments in the Massachusetts House of Representatives,

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[164] Massachusetts Commission Against Discrimination, "Workplace Discrimination Based on Your Genetics," Accessed April 14, 2026, https://www.mass.gov/info-details/workplace-discrimination-based-on-your-genetics.
[165] "An Act Relative To Insurance And Genetic Testing And Privacy Protection.," Acts of 2000 Chapter 254, Accessed April 14, 2026, https://malegislature.gov/Laws/SessionLaws/Acts/2000/Chapter254.
[166] Acts of 2000 Chapter 254, "An Act Relative To Insurance And Genetic Testing And Privacy Protection."

providing the perfect opportunity to make the bill more comprehensive in protecting genetic privacy before it becomes law.

If *Bill S. 2619* is amended to restrict genetic insurance discrimination and then passed into law, the residents of Massachusetts will be able to make informed insurance decisions based on their genetic information while insurance companies are not. The Massachusetts state government would not incur any costs from restricting genetic insurance discrimination, however, a portion of the profit of insurance companies will be transferred over to residents of Massachusetts. Since residents of Massachusetts will gradually become more aware of their risk of illness or mortality based on their genetics, they will be able to decide with greater certainty whether they derive benefit from buying insurance. Insurance companies would thus lose customers who are unlikely to be seriously ill or at risk of death, meaning that they lost a stable income source with low risk of high insurance payouts, and will instead gain more customers who are likely to need insurance payouts, which will cut into their profit margins. However, even with the exact amount of money that will be transferred from insurance companies to residents of Massachusetts being unclear, Massachusetts insurance companies are not in danger of losing profitability since they make a 12.3% profit on all insurance transactions, significantly outperforming the national average of 4.5%.[167] By implementing genetic insurance discrimination prevention laws, insurance companies will not significantly suffer financially and the residents of Massachusetts will have some of their financial burden lifted. Additionally, these preventative laws will stop insurance companies from exploiting even more money from residents of Massachusetts than is already happening in the status quo.

[167] "Report on Profitability by Line by State in 2023," National Association of Insurance Commissioners, April 2025. Accessed April 29, 2026, https://content.naic.org/sites/default/files/publication-pbl-pb-profitability-line-state.pdf

Establish biologically-specific compute thresholds for AI systems trained on genomic data

Bill S.2619 currently regulates genomic data as a category of sensitive data subject to enhanced consent and minimization requirements, but it does not impose AI-specific provisions on computational models trained on genomic sequences. This bill, thus, should be amended to add a definitional section addressing AI systems whose training data includes human genomic, transcriptomic, or proteomic sequences, thereby imposing a lower regulatory threshold than those applied to general-purpose AI systems.

Massachusetts should adopt a tired notification and audit requirement for genomic AI systems. For systems trained at or above e23 FLOP on datasets that include human genomic data, developers deploying such systems should be required to register with the Attorney General, and provide a model card disclosing training data sources, context window length, along with privacy risks. They should also submit to an annual third-party privacy audit which assesses memorization risk. This threshold would be consistent with the biological sequence threshold established in the executive order by the Biden administration, 14110, which stated that biological AI presents capability risks at lower compute scales than general-purpose models.

Regulate context window size and effective retrieval performance in genomic AI

As discussed earlier, the context window of a genomic AI model determines the biological scope of the models' understanding, going beyond the idea of a technical parameter. A model with a 1 megab-ase context window is much different from one limited to 4096 nucleotides for instance, because it can mark long-range regulatory interactions, chromatin architecture, and structural genomic features that the short context models do not have the capacity to represent. These are what make models like Evo 2 really promising for disease prediction, and variant effect scoring, but most importantly, they also amplify de-anonymization risk, because a model which attends to 1 million nucleotides simultaneously encodes far more individually identifying information than a short context model. Herein, we propose a second way to amend Bill S. 2619 to define a category of extended-context genomic AI systems, which should include any AI system processing human genomics sequences at a context length greater than 50,000 nucleotides. The bill should require an explicit disclosure of the stated context window and then the effective retrieval distance, which would be measured as the distance in nucleotides at which the model's attention-based retrieval accuracy falls below a specific threshold. Also, there should be a privacy impact assessment implemented to address the risk of individual re-identification through latent space probing and membership interference at certain context lengths. Ultimately, the Attorney General should be authorized to promulgate technical standards in consultation with Massachusetts institutions like Havard and MIT. These amendments still match the broader goal of S. 2619 to protect the privacy rights of Massachusetts residents, but now goes even further to acknowledge the growing class of privacy risks that are specific to long-range interaction modeling in genomic AI.

Implement a familial risk disclosure and enhanced consent requirement for AI use of genomic data

Massachusetts currently regulates and considers genetic information as highly sensitive under Massachusetts General Laws Chapter 111, Section 70G, requiring prior written informed consent for genetic testing and the disclosure of results[168]. However, this framework is based on

[168] Massachusetts General Laws, Chapter 111, §70G (Genetic Information Privacy). https://malegislature.gov/Laws/GeneralLaws/PartI/TitleXVI/Chapter111/Section70G

an individualized model of privacy, assuming that genetic data belongs only to the consenting person. As artificial intelligence (AI) becomes increasingly integrated into healthcare, particularly in genomic analysis and disease prediction, this assumption no longer reflects how genetic data is used or the scope of its impact.

To address this gap, the Massachusetts State Legislature should amend Chapter 111 to require a familial risk disclosure and enhanced informed consent process when genomic data is used in AI systems. This policy would require healthcare providers to explicitly inform patients, before consent, that their genomic data may reveal identifiable or predictive information about biological relatives when used in AI model development or large-scale data aggregation.

The enhanced consent requirement should include:

1. Disclosure that genomic data is partially shared with biological relatives and may expose familial health risks
2. Explanations of how AI systems may use genomic data to generate predictive insights beyond the individual patient
3. Address the potential for reidentification through data aggregation and familial matching
4. Ask for separate, documented consent specific to AI-related uses of genomic data

This recommendation responds to well-established limitations in genomic privacy. Unlike other forms of health information, DNA is inherently identifying and cannot be fully anonymized; multiple studies have shown that de-identified genomic data can be re-identified when combined with external datasets[169]. Additionally, genomic datasets frequently include familial relationships, and research has shown that genealogical inference methods can identify a substantial proportion of individuals in genomic datasets through relatives—underscoring the privacy risks created by shared genetics[170]. This relational nature of genomic data has been proven time and time again, as seen in the Golden State Killer investigation: law enforcement was able to identify a suspect by matching crime scene DNA to distant relatives in public genealogy databases

AI significantly amplifies these risks. These systems rely on large-scale genomic datasets, including federally funded programs like the NIH "All of Us" Research Program, which has sequenced hundreds of thousands of participants and generated millions of genetic variants for research use[171]. These systems aggregate and analyze genomic information across populations, enabling predictive modeling that extends beyond individual patients and increases the likelihood of re-identification through pattern recognition and data linkage.

Simultaneously, AI is increasingly becoming embedded in clinical care. The U.S. Food and Drug Administration has now authorized over a thousand AI-enabled medical devices, many of which depend on large datasets for training and validation[172]. This expansion means that genomic data is no longer used solely for individual diagnosis or independent research, but

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[169] Gymrek, M. et al. "Identifying Personal Genomes by Surname Inference." Science, 2013. Demonstrates re-identification of supposedly de-identified genomic data using external databases. https://www.science.org/doi/10.1126/science.1238282

[170] Erlich, Y. et al. "Identity inference of genomic data using long-range familial searches." Science, 2018. Shows that a large proportion of individuals in genomic datasets can be identified through relatives using genealogical methods. https://www.science.org/doi/10.1126/science.aau4832

[171] National Institutes of Health (NIH), All of Us Research Program Overview. Reports large-scale genomic sequencing and multi-hundred-million variant dataset creation. https://allofus.nih.gov

[172] U.S. Food and Drug Administration (FDA), Artificial Intelligence/Machine Learning (AI/ML)-Enabled Medical Devices List. Indicates over 1,000 authorized AI/ML-enabled medical devices as of recent updates. https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-aiml-enabled-medical-devices

increasingly serves as a foundational input for scalable AI systems used in clinics across populations.

Current policy proposals requiring AI disclosure in healthcare—including recent state legislative efforts on clinical AI transparency—primarily focus on whether AI is involved in treatment decisions, and if so, disclosing that information. However, they do not address how patient genomic data contributes to the development, training, or long-term functioning of AI systems, nor do they account for the broader familial implications of genomic data use.

This policy implementation would require the Department of Public Health to:

- Update consent guidelines and templates
- Provide standardized language for familial risk disclosure
- Integrate AI-specific genomic consent into existing clinical workflows.

The direct costs associated with this policy are minimal, as implementation would primarily involve updates to existing electronic health record systems and standardized consent documentation rather than the creation of new infrastructure and resources.

By recognizing the relational nature of genomic data and the expanding role of AI in healthcare, this policy ensures that informed consent in Massachusetts reflects individual autonomy and the broader privacy implications for relatives and long-term data use.

Codify Federated Learning as the Default Standard for State-Funded Multi-Institutional Genomic Research

The Massachusetts State Legislature ought to mandate that any research collaborations funded under the DRIVE Act (Bill H.4375), which establishes a $200 million Research Resilience Fund, utilize Federated Learning (FL) as the primary architectural standard for multi-omic analysis.[1] Current genomic research models often rely on centralizing sensitive data, an approach that creates persistent re-identification risks because genomic data is "inherently and permanently identifiable.”"[2,3] FL thereby offers a "privacy-preserving" alternative by shifting the paradigm from "moving data to models" to "moving models to data."[4]

Under an FL framework, raw genomic sequence data remains within the local firewalls of the originating institution, such as a rural Community Health Center or a Boston-based research hospital.[4] Only encrypted model updates such as gradients or weights are transferred to a central hub for aggregation into a global predictive model.[4] This architecture directly addresses the "Treatment Exception" loophole in HIPAA by ensuring that a patient’s entire "book of life" is never physically transferred to third-party AI developers without a clear, localized audit trail.[5] By embedding FL into the DRIVE Act’s funding requirements, the Commonwealth can catalyze innovation while upholding the high standards of Affirmative Consent established in Bill S.2516.[1,10]

[173] Maura Healey, “Bill H.4375: DRIVE Act,” The 194th General Court of the Commonwealth of Massachusetts, accessed April 21, 2026, https://malegislature.gov/Bills/194/H4375/BillHistory.
[174] W. Nicholson Price 2nd and I. Glenn Cohen, "Privacy in the Age of Medical Big Data," Nature Medicine 25 (2019): 37–43.
[175] Ellen Wright Clayton et al., "The Law of Genetic Privacy: Applications, Implications, and Limitations," Journal of Law and the Biosciences 6, no. 1 (2019): 1–36.
[176] Ying-Cheng Wu et al., "The Evolving Role of Artificial Intelligence in Medical Genetics: Advancing Healthcare, Research, and Biosafety Management," Genes 17, no. 1 (2025): 6.

Background and Case Study

The quantitative basis for FL in genomics rests on its ability to handle massive datasets expected to reach 40 exabytes in the next decade.[8] However, a critical technical challenge known as "client drift" must be addressed. Drift occurs when local datasets are non-IID (Independent and Identically Distributed), meaning genomic profiles in a rural Berkshire clinic may differ significantly from an urban Boston cohort.[7] If left unmanaged, the global model will naturally optimize for the "local majority" of the largest data nodes, which are typically wealthy, urban institutions, therefore creating a "weighted average of the wealthy" that reinforces health disparities.

Establish Minimum Privacy Thresholds and a Standardized "Policy Budget" for Genomic AI Models

To operationalize the privacy benefits of Federated Learning, the Massachusetts Department of Public Health (DPH), in coordination with the Office of Consumer Affairs and Business Regulation (OCABR), should establish standardized benchmarks for Differential Privacy (DP).[9,10] DP provides a mathematical guarantee of privacy by injecting "noise" into the model updates, ensuring that no individual's specific genetic markers can be inferred from the final aggregated AI model.[4]

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[177] U.S. Department of Health and Human Services, "Summary of the HIPAA Privacy Rule," 2013, https://www.hhs.gov/hipaa/for-professionals/privacy/laws-regulations/index.html.
[178] Dat Duong and Benjamin D. Solomon, "Artificial Intelligence in Clinical Genetics," European Journal of Human Genetics 33 (2025): 281–288.
[179] National Human Genome Research Institute, "Artificial Intelligence, Machine Learning and Genomics," January 12, 2022.
[180] "Case 1: An Act Modernizing Tobacco Control," Massachusetts Department of Public Health, 2019 Mass. Acts 133.

A central component of this regulation must be the definition of a privacy budget. A lower policy budget indicates stronger privacy but can result in reduced model accuracy, while a higher policy budget improves diagnostic precision at the cost of privacy. The Commonwealth may benefit significantly from convene a technical oversight committee to set specific policy budget benchmarks based on clinical utility:[181][182][183]

- High-Stakes Diagnostics: A slightly higher policy budget may be permitted for rare disease identification where high precision is a clinical necessity.[7]
- Population Health Screening: A lower, more restrictive policy budget should be mandated for broad screening applications where the risk of re-identification across large cohorts is higher.[2]
- Quantitative Benefits and Equity Considerations Mandating these thresholds ensures that the "Digital Shadow" gap is closed through technical guardrails.[3] Without a quantified privacy floor, AI models trained on genomic data remain vulnerable to "intelligent inference attacks" that bypass traditional anonymization.[4] Furthermore, to prevent model drift from harming underrepresented demographics, the state should implement incentive structures such as supplemental "Research Catalyst" grants for institutions that contribute high-quality, diverse genomic data from underrepresented populations.[1],4 This prevents the global model from being skewed by the ancestral biases currently prevalent in European-centric public databases.[4]

Implementation Strategy and Regulatory Oversight

[181] Michael O. Moore, "Bill S.2516: Massachusetts Data Privacy Act," The 194th General Court of the Commonwealth of Massachusetts, 2025, https://malegislature.gov/Bills/194/S2516.
[182] Andres X. Vargas, "Bill H.83: Massachusetts Data Privacy Protection Act," The 194th General Court of the Commonwealth of Massachusetts, 2023, https://malegislature.gov/Bills/194/H83.
[183] Michael O. Moore, "Bill S.2516," Section 1: Definitions (Large Data Holder), 2025.

Relevant Regulatory Bodies The DPH should serve as the primary clinical overseer, ensuring that AI-driven insights meet the FUTURE-AI principles (Fairness, Universality, Traceability, Usability, Robustness, and Explainability).[4,9] Simultaneously, the OCABR should manage the Data Broker Registry and enforce the Private Right of Action established in Bill H.83, allowing residents to seek liquidated damages of at least $15,000 per violation if privacy budgets are exceeded.[10,11]

Rollout and Compliance Implementation should occur in two phases. Phase I (Years 1-2) will require all DRIVE Act recipients to conduct a Data Protection Assessment and adopt FL for all data-sharing pilots.[1,10] Phase II (Years 3-5) will mandate that all Large Data Holders, which are defined in Bill S.2516 as entities with over $200M in revenue, comply with the state's policy budget benchmarks for any genomic AI tool deployed in clinical settings.[11]

Estimated Cost Analysis While the transition to FL requires significant infrastructure investment in specialized hardware (GPUs) and personnel, the long-term costs of data breaches and re-identification lawsuits far outweigh the initial setup.[4,8] By utilizing state-level funding from the DRIVE Act to subsidize the cost of "Privacy by Design" for rural CHCs, the Commonwealth can prevent a bifurcation of care and ensure a digitally ready workforce.[1,10]

Amend Chapter 111 of the General Laws to promulgate regulations governing the deployment of patient-facing artificial intelligence systems

In recent years, the Commonwealth of Massachusetts has taken steps to support the responsible development and deployment of artificial intelligence systems, including through the Massachusetts AI Strategic Task Force and the establishment of the Massachusetts AI Hub. At the same time, existing provisions of the General Laws, including M.G.L. c. 111, § 70G, recognize genetic and health-related information as sensitive and deserving of heightened protections. As artificial intelligence systems are increasingly integrated into diagnostic support, risk stratification, and treatment selection, there exists a need to clarify the standards under which such systems may be deployed in patient-care settings. In particular, current law does not specify the degree to which developers or deployers of clinical AI systems must demonstrate that model outputs are grounded in clinically relevant features rather than spurious correlations or confounding variables. Massachusetts tort law explicitly precludes "unfair and deceptive" practices in commercial (and by extension, health) contexts. An analogous view should be taken with regard to AI usage in genomic analysis.

Legislature direct the Department of Public Health, in consultation with the Health Policy Commission and relevant clinical stakeholders, should promulgate regulations establishing interpretability and validation requirements for artificial intelligence systems used in patient-facing healthcare contexts. These requirements should be calibrated to the level of clinical risk, the architecture of the model, and the extent to which the system directly informs patient care decisions, while preserving flexibility for research and innovation.

As used in this proposal, "patient-facing artificial intelligence system" shall mean any computational model whose outputs are used to inform diagnosis, prognosis, treatment selection, or the communication of clinically relevant information to a patient or provider acting on behalf of a patient. This definition is intended to distinguish such systems from research-only tools, which may be subject to institutional review but not to the same regulatory requirements.

The proposed regulatory framework would amend Chapter 111 of the General Laws by adding a new section requiring that the Department promulgate regulations governing the deployment of patient-facing artificial intelligence systems. Such regulations should include, but not be limited to, the following provisions.

1) The regulations should establish a baseline requirement of demonstrated clinical validity and robustness to confounding variables. Developers or deployers of patient-facing systems must provide evidence that model performance is not primarily driven by spurious correlations, dataset artifacts, or proxies unrelated to the underlying clinical phenomenon. Acceptable forms of evidence may include external validation on independent datasets, subgroup analyses across demographic and clinical variables, and controlled perturbation or ablation studies demonstrating that model outputs change appropriately in response to clinically meaningful input variation. The purpose of this requirement is not to mandate complete mechanistic understanding of model internals, but rather to ensure that model efficacy reflects genuine clinical signals rather than coincidental or biased features of the training data.
2) Regulations should differentiate requirements based on model architecture and scope, recognizing that not all artificial intelligence systems present the same interpretability challenges.
    a) Transformer-based sequence models used for narrowly defined predictive tasks such as TFBS prediction, promoter region prediction, or splice site recognition may be evaluated using established techniques including feature attribution, counterfactual testing, and performance benchmarking across clinically relevant subgroups. In such cases, the Department may

deem a system compliant where the developer demonstrates that the model reliably responds to known clinical or biological features and does not rely on identifiable confounders.

b) Broader foundation models such as large-scale multimodal systems or long-context genomic models capable of predicting multiple outputs across diverse conditions may require additional documentation and testing. For such systems, regulations should require evidence that model outputs remain stable and clinically meaningful across variations in input context, as well as documentation of known limitations in specific populations or care settings. However, the regulations should not require exhaustive mechanistic interpretability where such understanding is not technically feasible, provided that the system demonstrates consistent and validated clinical performance. The intent here is not to bring about a "chilling effect" on AI innovation, but rather to ensure that general-purpose genomic foundation models (such as Evo and Evo-2) have outcomes grounded in known biological signals and remain sufficiently deterministic to satisfy long-term patient health needs.

3) The regulations should impose heightened requirements on systems with generative or design capabilities that may influence clinical decision-making in less transparent ways. Where a model is capable of generating novel clinical recommendations, treatment options, or biological hypotheses, the Department should require clear documentation of the intended use, constraints on

deployment, and evidence that generated outputs are subject to appropriate human oversight. In such cases, interpretability may be supplemented by procedural safeguards, including clinician review and audit trails, rather than relying solely on model-internal explanations. SB2632 already limits the autonomous use of AI systems in healthcare settings. This is a standard that should be continued. Model outcomes that are generative rather than predictive or diagnostic should trigger a stricter review from physicians or relevant technicians.

4) Regulations should also include a requirement that all patient-facing artificial intelligence systems be accompanied by a standardized disclosure or documentation file, to be made available to deploying institutions and, where appropriate, to patients. Such documentation should include a description of the model architecture, intended use cases, training data characteristics, validation procedures, known limitations, and any identified risks of bias or confounding. The Department may develop and make available a standardized format for such documentation, similar to existing requirements for clinical labeling or device documentation.

5) Regulations should explicitly provide that research and pre-clinical development activities are subject to a more flexible standard, provided that such systems are not used to inform patient care decisions. Academic researchers and developers operating within institutional review board (IRB)-approved protocols should not be required to meet the full set of interpretability and validation requirements until such systems are deployed in patient-facing contexts. This distinction is intended

to avoid imposing undue burdens on early-stage innovation while ensuring that systems entering clinical use meet appropriate standards of reliability and safety.

6) The regulations should incorporate a **risk-based enforcement framework**, under which systems used in higher-stakes clinical decisions such as cancer diagnosis, surgical planning, or medication selection are subject to more stringent validation and documentation requirements than systems used for administrative support or low-risk clinical assistance. The Department may coordinate with federal regulators, including the U.S. Food and Drug Administration, to align state requirements with existing guidance on software as a medical device, while retaining authority to address gaps specific to the Commonwealth.
7) Finally, the regulations should include provisions to periodically update interpretability and validation standards in light of evolving model architectures and scientific understanding. The field of artificial intelligence is rapidly advancing, and regulatory requirements should be sufficiently flexible to accommodate new methods for demonstrating robustness and clinical validity. To that end, the Department may convene an advisory group of technical experts, clinicians, and patient representatives to review and recommend updates to the regulatory framework on a regular basis.

In implementing these recommendations, the Commonwealth should seek to balance the need for patient safety and trust with the importance of maintaining Massachusetts as a leading center for biomedical innovation. By focusing on demonstrable clinical validity, robustness to confounding variables, and transparency in deployment—rather than mandating complete mechanistic understanding of complex models—the proposed framework aims to ensure that artificial intelligence systems used in patient care are both effective and responsibly governed, without imposing unnecessary barriers to innovation.

Implement Massachusetts House Bill H.4807: An Act relative to updating the security of personal information:

Massachusetts should implement Massachusetts House Bill H.4807 to secure Massachusettsans' private health and general data. Massachusetts already exhibits significantly more rigorous security laws than most of the rest of the country. This act increases protections to new types of data and imposes far more stringent controls on the use and regulations surrounding this data.

The first part of this bill updates Massachusetts Statutory Law Chapter 93H. This chapter regulates how private data may be handled in Massachusetts, and what actions must be taken in situations for which errors occur while handling the data. Bill H.4807 updates this language to include new types of Private Health Data that was not previously covered. This legislation includes credit card information, biometric identifiers, and even neural data.

I. Background Research & Case Study

HIPAA was designed in a time when none of the newly protected identifiers were readily available. Since then, HIPAA has failed to update to reflect the changing healthcare landscape. Breaches of private health information mentioned above can lead to devastating consequences, and American taxpayers are not thoroughly protected against these losses. Biometric data breaches have become increasingly common in particular. Many critical online systems and portals include biometric identifiers to secure information. When this data is stolen, criminals get permanent access to information that is being used more frequently to secure the most sensitive consumer information. This often includes two- factor authentication services as well, which could make accessing even the most well-protected information relatively easy for the motivated cybercriminal.

This stolen data can be used in a variety of ways, and all can be aided by advancing AI models. Deepfakes are particularly concerning. In these cases, hackers will steal biometric information and create fake videos using your likeness. These videos often put the victim in compromising situations or are used to extort loved ones. These schemes are often aided by AI. Without protections on biometric information, holders of this information have no incentive to protect your data from potential breaches.

Amend Regulation *101 CMR 20.00*: Health Information Exchange to Establish Mandatory AI Auditing for Genomic Data Systems in Massachusetts

Massachusetts Executive Office of Health and Human Services (EOHHS) should amend the Regulation 101 CMR 20.00: Health Information Exchange – that is, the Mass HIway Regulations – to require an artificial intelligence (AI) screening policy for all genomic uses and transfer.[184] Currently, the Health Information Exchange regulates the transfer of genomic data, but only requires hospitals and healthcare centers to connect to the system, so all outside organizations using genomic data, such as labs or educational institutions, are not required to follow the same rules. The amendment can be enforced by EOHHS, which currently upholds regulation 101 CMR 20.00. Furthermore, the act should keep a detailed log of all inputs and outputs, as well as the access and transfer of data, to increase the efficiency and effectiveness of

[184] "101 CMR 20.00: Health Information Exchange," Mass.gov, accessed April 27, 2026, https://www.mass.gov/regulations/101-CMR-2000-health-informatiprevalenton-exchange.

the audits. Finally, the act ought to assign risk scores to the data that align with clearance based on identifiability and sensitivity of the data, which I will elaborate on in a later recommendation.

Background Research & Case Study

Artificial intelligence (AI) use in genomic data analysis has become increasingly prevalent, but as the use of machine learning (ML) methods has increased, the safety regulations have not kept up with the speed of development.[185] This inequality has created gaps in genomic data safety, which has led to an increase in data leaks. Approximately 40% of organizations report an AI-related privacy incident, and, in 2025 alone, a 68% increase in AI-related data incidents was observed nationwide.[186] While Massachusetts does have safety regulations in place, such as *101 CMR 20.00: Health Information Exchange,* which regulates the transfer of genomic data, these systems are not perfect. Mass HIway requires compliance from providers, and it is only required …

not required in all circumstances. The providers required to connect to the system are "Acute Care Hospitals, Large and Medium Medical Ambulatory Practices, and Large and Small Community Health Centers," as well as any organizations that the Massachusetts Executive Office of Health and Human Services includes.[187] Therefore, in any outside research involving genomic data, there does

[185] Dara Mahintaj et al., "The Transformative Role of Artificial Intelligence in Genomics: Opportunities and Challenges ," ScienceDirect, 2025, https://www.sciencedirect.com/science/article/abs/pii/S2452014425001876.
[186] "AI Data Privacy Statistics & Trends 2025," Protecto AI, September 4, 2025, https://www.protecto.ai/blog/ai-data-privacy-statistics-trends/?utm.
[187] "Hiway Regulations Faqs," The Mass HIway, January 2022, https://www.masshiway.net/node/90.

not exist a comprehensive mandatory framework that regulates the transfer and use of data. Thus, safety systems for generative AI used for genomic data need to be updated with more regulations so that the Massachusetts healthcare system can efficiently utilize AI tools without significant risk of private data leaks.

Integrate Embedded Safety Requirements into HIPAA for AI-Driven Genomic Data Systems

Moreover, the federal Health Information Privacy and Accountability Act of 1996 (HIPAA) should be amended to require embedded safety standards for all AI that uses, transfers, or stores genomic data. Currently, its purpose is to regulate the handling of data through external oversight. Instead, HIPAA should require that safeguards be implemented directly into the system's architecture to minimize safety risks at the point of processing. This includes prevention against unnecessary data movement and re-identification. While this structure reduces the need for external regulation, routine audits should still be performed to detect any problems. It should be noted that implementing embedded systems could limit the visibility of the model's behavior. However, this system would prevent many unnecessary data leaks from occurring.

HIPAA regulates the use and disclosure of 18 specified identifiers, but genomic data is not listed as an identifier.[188] However, genomic data has been proven to be a major identifier for individuals. One specific study showed how technology can be used to recreate profiles from

[188] Jennifer Kulynych and Henry T. Greely, "Clinical Genomics, Big Data, and Electronic Medical Records: Reconciling Patient Rights with Research When Privacy and Science Collide," *Journal of Law and the Biosciences* 4, no. 1 (2017): 94–132, https://doi.org/10.1093/jlb/lsw061such as HIPAA, or approved online healthcare platforms, they were not designed for the immense volume.

genomic data, and they acknowledged that re-identification would increase.[189] It has already been a few years since that study was conducted, and now it has been shown that, through genetic genealogy databases, a majority of the population in America can be identified from their DNA.[190] These privacy breaches are occurring even with current regulations such as HIPAA or Mass HIway, and other privacy laws in Massachusetts.

Massachusetts General Law Chapter 111 Section 70G: Genetic information and reports protected as private information; prior written consent for genetic testing states that any genomic data that is identifiable from the genes can not be made public or disclosed without written consent from the patient. Furthermore, it says that without written consent, a patient cannot be tested for genetic information nor have their results disclosed to any person. However, the genetic genealogy databases prove that even opting out of data sharing or making your health information public does not guarantee your safety. Expanding HIPAA to include embedded privacy systems would reduce these concerns.

[189] Jennifer Kulynych and Henry T. Greely, "Clinical Genomics, Big Data, and Electronic Medical Records: Reconciling Patient Rights with Research When Privacy and Science Collide," *Journal of Law and the Biosciences* 4, no. 1 (2017): 94–132, https://doi.org/10.1093/jlb/lsw061.
[190] Megan Molteni, "Genome Hackers Show No One's DNA Is Anonymous Anymore," Wired, October 11, 2018, https://www.wired.com/story/genome-hackers-show-no-ones-dna-is-anonymous-anymore/#:~:text=In%202013%2C%20a%20young%20computational,of%20anonymized%20biomedical%20genetic%20data.